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Bayesian Clustering

by

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Summary

KPMG Nederland has provided solutions in Audit & Assurance, Advisory, and Tax & Legal for over 100 years [14]. The Advisory team offers expert guidance on managing risk and improving business performance to support future growth. As part of this mission, the Advanced Analytics & Big Data team undertakes projects that leverage data and AI to design and implement scalable solutions. The team invests in *research and innovation*, enabling them to remain at the forefront and deliver impactful solutions across diverse sectors.

Clustering is a method of *unsupervised machine learning* used to uncover inherent patterns in complex, high-dimensional datasets—insights that are critical when advising clients in industries such as finance, healthcare, government, retail, and more. Common applications include image processing, customer segmentation, and anomaly detection.

The integration of *Bayesian* methods in clustering provides a principled framework to handle uncertainty by incorporating prior knowledge into the model. As a result, these methods enable organizations like KPMG to make more informed predictions and decisions. They produce qualitative outputs, such as distributions over possible outcomes, allowing decision-makers to quantify uncertainty using appropriate measures.

Bayesian Clustering is a research master’s project conducted within the context of an internship at KPMG Nederland, aimed at real-world applicability. Consequently, challenges such as *Performance*, *Scalability*, *Interpretability*, and *Robustness* must be effectively addressed. The theoretical foundation builds upon the concept of Bayesian Clustering for general k -groups, originally proposed by E. Belitser in [3], and is further developed in Chapters 1 and 2. A new class of algorithms along with a ready-to-use Python package [bayesiancl], and their performance evaluation, are detailed in Chapters 3 and 4.

Title: Bayesian Clustering

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Contents

1	Introduction	9
1.1	Overview of Clustering Approaches	9
1.2	Partition Space and Notation	11
1.3	Bayesian Inference	12
1.4	Hierarchical Modeling in Bayesian Framework	13
1.5	From Sparsity in " <i>Signal + noise</i> " model to Clustering	15
2	Bayesian Clustering Model	18
2.1	Clustering Structure: A Linear Subspace Approach	18
2.2	Bayesian Analysis for known partition	19
2.3	Bayesian Analysis for unknown partition	20
2.4	Binary Clustering	21
2.4.1	Prior in Product-form	22
2.4.2	Cardinality-based prior	25
2.5	k -Groups Clustering	27
2.5.1	Prior in Product-form	28
2.5.2	Cardinality-based prior	31
2.6	Prediction	34
2.6.1	Point-wise Posterior	35
2.7	Example: Normal Data	40
2.8	Quality Measures for Performance Assessment	41
2.8.1	Adjusted Rand Index	41
2.8.2	Variation of Information	43
2.8.3	Set Matching criterion	45
3	Simulation study	48
3.1	Dataset Design	48

3.1.1	True Partition and Cluster Means	49
3.2	Simulation Study - Results	51
3.2.1	Effect of the Number of Clusters	51
3.2.2	Effect of the Separation Parameter δ	52
3.2.3	Performance Comparison with KMeans and DBSCAN	53
4	Real data and application	56
4.1	The Iris Flower Dataset	56
4.2	The Iris dataset - Results	57
4.2.1	Prior Mean Selection	57
4.2.2	Number of clusters	58
4.2.3	Model assumptions	59
4.2.4	Distribution of Clusters	61
5	Conclusion	63
5.1	Conclusion	63
5.2	Future Work	65
6	Bibliography	66
Appendix		69
A.1	Proofs: Cardinality-based prior as a probability distribution	69
A.2	Code Snippets and Explanations	70
A.3	Additional figures and tables: Iris Dataset	70
A.4	Code availability	71

1 Introduction

In *Machine Learning*, the grouping of objects or data points into "categories" is addressed by two major fields; in supervised learning where labels are provided, *Classification* approaches are used to tackle the already mentioned task, while, *Clustering* is a grouping technique of unsupervised learning where the labels are little or not available in the given data. Therefore, in general, Clustering is more difficult and challenging than Classification. The objective of such analysis; *Clustering analysis* is to develop an automatic algorithm that will discover the natural groupings in the unlabeled, unsupervised data [13].

Moreover, a *Cluster* is vaguely defined as the set of objects with a higher degree of similarity between them than with the objects outside the set [17]. In order to argue the similarities and dissimilarities between data points, suitable measures should be used which consequently give rise to different *clustering methods*.

1.1 Overview of Clustering Approaches

The underlying clustering structure of the various available clustering algorithms serves as a key factor in differentiating between them. Their structure comprises of the choice of the objective function, the probabilistic generative models and the heuristics. Three central and widely used families of clustering approaches are: *Partition-based Clustering*, *Model-Based Clustering*, and *Density-based Clustering* [17].

Partition-based clustering methods, such as K-Means, are known for their efficiency and scalability. They are easy to interpret due to *exclusive cluster membership*, meaning each data point must belong to exactly one cluster. On the other hand, they need to predefine the number of clusters and they are sensitive to initialization of the centroids in the loss function, Table 1.1. In addition, they struggle with non-convex clusters, noise, and outliers.

Density-based clustering defines clusters as dense regions in the data space, separated by regions of lower point density. Exemplified by the *Density-Based Spatial Clustering of Applications with Noise* (DBSCAN), it excels in identifying clusters of arbitrary shapes and is robust to noise and outliers. It doesn't require specifying the number of clusters but parameters defining the radius of neighborhood around a point and minimum neighbors within that radius, which are usually domain specific, require careful parameter tuning. As a result DBSCAN can be computationally intensive. Furthermore, it is sensitive to the curse of dimensionality due to its reliance on Euclidean distance measures.

Model-based clustering, such as Gaussian Mixture Models, offers flexibility by modeling data with probabilistic distributions and handling overlapping clusters effectively. It shares the weaknesses of the previously mentioned methods in terms of converging to local optima based on seed assignment and the requirement of specifying the number of clusters a priori. Additionally, the optimization criteria can theoretically take infinite iterations to reach convergence. While it provides valuable interpretability through soft clustering: each point gets a probability distribution over clusters, it can be computationally demanding and less scalable, especially for large datasets.

Partition-Based The method assumes initial centroids. The objective is to minimize the mean squared error loss function with respect to the centroids, which are usually, either the mean or the median of the points. Each centroid represents each cluster [22]. $\sum_{j=1}^k \sum_{i=1}^n \ x_i^{(j)} - c_j\ ^2$ where $\ x_i^{(j)} - c_j\ ^2$ is chosen as the distance measure of point x_i from centroid c_j.	Model-Based A GMM assumes that the data points $x_1, x_2, \dots, x_n$ are generated from a mixture of K Gaussian distributions each with its own mean vector μ_k and covariance matrix Σ_k. $p(x_i \Theta) = \sum_{k=1}^K \pi_k \cdot \mathcal{N}(x_i \mu_k, \Sigma_k)$ The objective is to maximize, $L(\theta) = \sum_{i=1}^N \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i \mu_k, \Sigma_k) \right)$
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Table 1.1: Partition-Based and Model-Based Clustering

Ultimately, selecting the most appropriate method depends on the specific charac-

teristics of the data and the goals of the analysis, taking into account trade-offs between efficiency, robustness, scalability, and complexity — including challenges such as predefining the number of clusters or other parameters, sensitivity to initialization as well as convergence.

This study employs two model-based clustering methods, which will be evaluated in comparison with partition-based (KMeans) and density-based (DBSCAN) methods in Chapters 3 and 4. Before introducing the method in detail, we begin by establishing some preliminary notation and concepts.

1.2 Partition Space and Notation

Let $[n]$ be the set $\{1, \dots, n\}$. A single clustering can be viewed as a partition of the indices $[n]$ of n data points into k disjoint sets. Following the notation in [23], a single partition can be represented either by an ordered collection of sets $I = (I_1, \dots, I_k)$, where each set $I_j \subseteq [n]$ contains the indices of the data points assigned to the j^{th} cluster, and k denotes the number of clusters in a sample of size n . Alternatively, the partition can be represented by a label vector $\mathbf{c} = (c_1, \dots, c_n)$, where $c_i = j \in [k]$ if the i^{th} data point belongs to the j^{th} cluster.

The partition space $\mathcal{I}_k$ denotes the set of all possible partitions of n points into k disjoint groups:

$$\mathcal{I}_k = \{I = (I_1, \dots, I_k) : \cup_{i=1}^k I_i = [n] \text{ and } I_i \cap I_j = \emptyset \text{ for all } i, j \in [k] : i \neq j\}. \quad (1.1)$$

Following the notation introduced in [4] and [6], let $|I_i|$ be the cardinality of the set I_i , $i \in [k]$. For a given partition $I = (I_1, \dots, I_k) \in \mathcal{I}_k$, let the cardinality-vector $\mathbf{m} = (|I_1|, \dots, |I_k|)$ represent the cardinalities of the respective sets $I_1, \dots, I_k$. Note that, according to (1.1), it holds that, $|\mathcal{I}_{k-1}| \leq |\mathcal{I}_k|$, for $k \in [n]$, meaning that empty clusters are possible and, therefore, $|\mathcal{I}_k| = k^n$.

Additional notation

For simplicity, we often omit the dependence of probability measures (e.g., P_θ) on σ and n . Let $\mathbb{E}(\cdot)$ and $\mathbb{V}(\cdot)$ denote expectation and variance, respectively. For sets S and S_0 , the set difference is defined as $S \setminus S_0 = \{s \in S : s \notin S_0\}$. We use the notation $[n]_0 = \{0\} \cup [n]$ and $\mathbb{N}_0 = \{0\} \cup \mathbb{N}$. Given a subset $S \subseteq [n]$ and a vector $v = (v_i, i \in [n]) \in \mathbb{R}^n$,

define the complement $S^c = [n] \setminus S$ and the subvector $v_S = (v_i, i \in S) \in \mathbb{R}^{|S|}$. For a partition $I = (I_1, \dots, I_k) \in \mathcal{I}_k$, define the corresponding subvectors as

$$v_1 = (v_i, i \in I_1), \quad \dots, \quad v_k = (v_i, i \in I_k).$$

The Bayesian model developed in this work is based on the structured clustering representation via partitions of the index set $[n]$ introduced in Section 1.2. In this framework, partitions act as latent variables capturing structural features like clustering or sparsity in the parameter vector $\theta \in \mathbb{R}^n$. To properly construct such a model, it is important to first outline the core concepts of Bayesian inference as well as hierarchical Bayesian modeling, presented in the following section.

1.3 Bayesian Inference

The frequentist view defines the probability of an event as the proportion of times that the event occurs in a sequence of possibly hypothetical trials. [2]

In the frequentist approach one will determine the corresponding parameter (i.e. mean) through the marginal likelihood or the method of moments. In the Bayesian framework, instead for picking fixed values, the parameters of interest are being treated as *random variables* which are generated by the so-called *prior* distribution.

The prior distribution expresses one's belief for each $\theta \in \Theta$ which differs from the frequentist approach; probability of an event is not approximated by repetition of the same experiment again and again.

Bayesian inference is inherently conditional on a specific observed sample. This approach can be vague in cases where the belief we hold about the underlying distribution is little or non existent.

Consider the random sample $X_1, \dots, X_n \sim f(x|\theta)$ where f is a density (w.r.t. some σ dominating measure) and our belief on the prior $\theta \sim \pi(\theta)$, the same dominating measure for all $\theta \in \Theta$. Intuitively this means that, θ is more likely to be around certain values and less likely around others. After those random variables have been realized; the observed data, $x_1, x_2, \dots, x_n$ are drawn from distribution $f(x|\theta)$. The probability of observing θ given $x_1, x_2, \dots, x_n$ according to *Baye's Theorem*, namely, the posterior dis-

tribution $\pi(\theta|x)$ is defined as,

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int_{\Theta} f(x|\theta)\pi(\theta)d\theta}, \quad (1.2)$$

where $\int_{\Theta} f(x|\theta)\pi(\theta)d\theta$ is the marginal distribution. Note that $\int_{\Theta} f(x|\theta)\pi(\theta)d\theta$ is independent of θ and acts as a normalizing constant, leading to,

$$\pi(\theta|x) \propto f(x|\theta)\pi(\theta).$$

In other words, $\pi(\theta|x)$ is the *a posteriori* belief one holds for the nature/distribution of the parameter of interest θ after observing the given data.

1.4 Hierarchical Modeling in Bayesian Framework

Based on the Bayesian inference framework introduced earlier, the Bayesian model can be described by a two-step generative algorithm that produces the data x and a parameter θ as follows,

Step 1: Generate θ according to $\pi(\theta)$,

Step 2: Given θ , generate x according to $f(x|\theta)$.

Here, x is the *observed* data whereas θ is an *unobserved* parameter drawn from the prior distribution $\pi(\theta)$, [21]. In case the unobservable, θ depends on additional unobserved variables beyond the parameter itself, another hierarchy level is introduced, and the generative model involves more than two steps. The generative algorithm then extends to,

Step 1: Generate α according to λ_{α} ,

Step 2: Given α , generate θ according to $\pi_{\alpha}(\theta)$, (1.3)

Step 3: Given α, θ , generate x according to $f(x|\theta)$.

In this hierarchy, every unobservable receives a distribution and after observing the data the distributions are updated to their conditional distributions given the data.

Theoretically, one can include as many hierarchy levels as necessary.

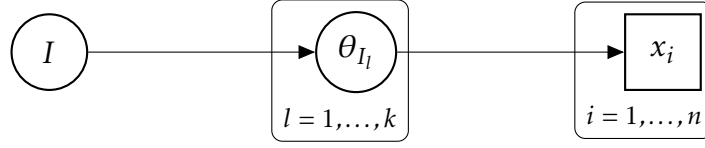


Figure 1.1: Hierarchical Bayesian clustering: I partitions data into groups, each with parameter θ_{I_l} generating x_i .

Example 1.1. In the context of clustering, we assume a two-level hierarchical structure as described in (1.3), involving the following random variables: the unobserved partition $I \in \mathcal{I}_k$, which is drawn first; followed by the unobserved parameter θ , drawn conditional on I ; and finally, the observed data points x , drawn given θ , as depicted in Figure 1.1. The resulting generative algorithm is as follows,

$$\begin{aligned}
 \text{Step 1:} \quad & I \text{ generated from } \lambda_I \\
 \text{Step 2:} \quad & \theta \text{ generated from } \Pi_I \\
 \text{Step 3:} \quad & x \text{ generated from } P_\theta
 \end{aligned} \tag{1.4}$$

where λ_I and Π_I denote prior distributions on I and θ respectively, for the model.

The aim of the following section is to establish how sparsity in high-dimensional parameters can be described using linear subspaces defined by partitions of the index set $[n]$. The idea of using a family of linear spaces to model general structures in high-dimensional parameter spaces originates from Belitser and Nurushev [5]. This approach generalizes traditional notions of sparsity to allow for multi-level structures where groups of parameters share common values. By formulating sparsity through these partition-based linear spaces, we create a foundation for the Bayesian hierarchical model to effectively represent clustering within the parameter θ .

1.5 From Sparsity in "*Signal + noise*" model to Clustering

Suppose that we observe a vector $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ such that

$$X_i = \theta_i + \epsilon_i, \quad i = 1, \dots, n \quad (1.5)$$

for independent random errors ϵ_i and an unknown vector of interest $\theta = (\theta_1, \theta_2, \dots, \theta_n)$, $\theta \in \mathbb{R}^n$. It is further assumed that the errors; $\mathbb{E}_\theta(\epsilon_i) = 0$ and $\mathbb{V}_\theta(\epsilon_i) = \sigma^2$, $i \in [n]$, where σ is the known noise intensity. Thus, θ is a vector of means [7].

The traditional characterization of sparsity is provided by the concept of *nearly black vectors*. In this setting, sparsity can be defined in terms of the ℓ_0 -ball and has just one known sparsity level which is, by default, zero. This is given by,

$$\ell_0(p) = \{\theta \in \mathbb{R}^n : \#\{i \in \{1, \dots, n\} : \theta_i \neq 0\} \leq p\},$$

where p is a fixed number of non-zero entries [1]. Taking into account the model (1.5) along with the traditional sparsity structure, it follows that $\theta \in \ell_0(p)$, indicating that θ has only a small proportion of nonzero coordinates (typically, $p \ll n$) and can, thus, be regarded as a sparse vector. The sparsity class can be expressed by the linear subspaces,

$$\mathbb{L}_0 = \{\theta \in \mathbb{R}^n : \theta_i = 0, \forall i \in I_1^c\}, \quad I = (I_1, I_1^c) \in \mathcal{I}_2.$$

Allowing the value of the sparse cluster to be unknown [8] the vector of interest becomes $\theta = (\theta_{I_1}, \theta_{I_1^c})$, where θ_{I_1} and $\theta_{I_1^c}$ are sub-vectors according to the introduced notation in 1.2 or, equivalently,

$$\mathbb{L}_{\mathcal{I}_2} = \left\{ \theta \in \mathbb{R}^n : \theta_i = \theta_j, \forall i, j \in I_1^c \right\}, \quad I \in \mathcal{I}_2. \quad (1.6)$$

Extending the above traditional one-level sparsity to namely the multi-level sparsity as in [6], it assumed that θ consists of some known values $\alpha_1, \dots, \alpha_k$, ($k < n$). By allowing the sparsity level to be some *unknown fixed constants*, the multi-level sparsity structure is generalized to sparsity defined by linear subspaces of the form,

$$\mathbb{L}_{\mathcal{I}_{k+1}} = \left\{ \theta \in \mathbb{R}^n : \theta_j = \theta_{j'}, \forall j, j' \in I_l, \forall l \in [k] \right\}, \quad (I_0, I_1, \dots, I_k) \in \mathcal{I}_{k+1}. \quad (1.7)$$

In this setting the hyperparameter is of the form, $\theta = (\theta_{I_0}, \theta_{I_1}, \dots, \theta_{I_k}) \in \mathbb{R}^n$, where θ_{I_0} is the signal class and $\theta_{I_1}, \dots, \theta_{I_k}$ are the unknown sparse classes.

In this thesis, we specifically focus on the clustering structure, which can be modeled using the family of linear spaces introduced in equation (1.7)

2 Bayesian Clustering Model

This chapter builds on the notes by E. Belitser [3], which introduce two types of priors: the product-form prior and the cardinality-based prior. For both, the work presents posterior computations using the “polynomial trick,” a key technique behind a clustering method based on point-wise posterior probabilities. We provide further explanation of this methodology and extend both prior types to the k -groups setting. Assume the following observation model, $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ such that,

$$X \sim P_\theta = \bigotimes_{i=1}^n P_{\theta_i}, \quad (2.1)$$

where $\theta = (\theta_1, \dots, \theta_n) \in \mathbb{R}^n$ is a high-dimensional unknown parameter. Thus the data distribution P_θ is a product measure.

Since, in general, we have n observations $(X_1, \dots, X_n)$ and n parameters $(\theta_1, \dots, \theta_n)$, reasonable inference on θ is impossible unless some structure on θ is imposed. In this thesis, we are concerned with a *clustering structure* on parameter θ .

2.1 Clustering Structure: A Linear Subspace Approach

The clustering structure on θ can be seen as an extension of the multi-level sparsity model described in (1.7), where the signal class is omitted. Specifically, a k -group clustering structure for θ is modeled by assuming $\theta \in \mathbb{L}_I$, for some $I \in \mathcal{I}_k$, where

$$\mathbb{L}_I = \{ \theta \in \mathbb{R}^n : \theta_i = \theta_j, \forall i, j \in I_l, l \in [k] \}, \quad I \in \mathcal{I}_k \quad (2.2)$$

where, the family of all partitions of $[n]$ in k groups, is given by,

$$\mathcal{I}_k = \left\{ I = (I_1, \dots, I_k) : \bigcup_{i=1}^k I_i = [n], I_j \cap I_{j'} = \emptyset, \forall j, j' \in [k] : j \neq j' \right\}.$$

Using the linear subspaces approach, the task is transformed into grouping the coordinates of θ into k distinct groups using the observed data $X \sim P_\theta$, $\theta \in \mathbb{L}_I$ and $I \in \mathcal{I}_k$. The inference on the underlying clustering $I_0 = I_0(\theta)$ of the “true” parameter θ consists of constructing a data-dependent clustering procedure $\hat{I} = \hat{I}(X)$ that aims to reproduce the “true” $I_0(\theta)$ as close as possible. Note that, I_0 is not unique; however, we are particularly interested in the one with the lowest complexity—in our case, the one with the smallest dimension. To assess the quality of the procedure $\hat{I}$, we will later introduce several quality measures $R(\hat{I}, I)$.

2.2 Bayesian Analysis for known partition

To formulate the Bayesian Clustering model with a known partition I , we focus on the three key components of Bayesian analysis introduced in Section 1.3; *the prior on θ , the marginal distribution for X and the posterior for θ given X* .

Lemma 2.1. *Let the data $X \sim P_\theta$ be according to the model (2.1), with the assumed structure on θ , $\theta \in \mathbb{L}_I$, where $\mathbb{L}_I$ is defined by (2.2) for a known partition $I = (I_1, \dots, I_k) \in \mathcal{I}_k$. Assume the prior $\theta \sim \Pi_I$, such that $\Pi_I = \bigotimes_{i \in [n]} \Pi_{I,i} = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l,i}$. Then the marginal distribution for X is given by, $P_{X,I} = \bigotimes_{i \in [n]} P_{X,I,i}(x_i) = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i)$ where,*

$$P_{X,I,i}(x_i) = \int P_{\theta_i}(x_i) d\Pi_{I,i}(\theta_i) = \int P_{\theta_i}(x_i) \Pi_{I_l^*,i} d\theta_i, \quad (2.3)$$

given $i \in I_{l^*}$ for some $l^* \in [k]$ and the posterior for θ is given by,

$$\Pi_I(\theta | X) = \bigotimes_{i \in [n]} \Pi_{I,i}(\theta_i | X_i) = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l,i}(\theta_i | X_i)$$

where,

$$\Pi_{I,i}(\theta_i | X_i) = \frac{\Pi_{I,i}(\theta_i) P_{\theta_i}(X_i)}{P_{X,I,i}(X_i)}, \text{ and } \Pi_{I_l^*,i}(\theta_i | X_i) = \frac{\Pi_{I_l^*,i}(\theta_i) P_{\theta_i}(X_i)}{P_{X,I_l^*,i}(X_i)}.$$

Proof. The posterior expression follows directly from the Bayesian Theorem (1.2). $\square$

2.3 Bayesian Analysis for unknown partition

Naturally, the objective of clustering is to identify an optimal grouping of the observations (based on some criteria). Therefore, the partition I should be considered unknown a priori. To account for this uncertainty, we assign a prior over the space of all possible partitions, $\mathcal{I}_k$. This introduces an additional hierarchical level to the model, broadening the scope of the analysis to include inference on the posterior of I given the data X , as previously described in the hierarchical model by (1.4).

Definition 2.2 (Bayesian Clustering Model). Let λ be a prior on the partition space $\mathcal{I}_k$, such that $I \sim \lambda_I$ and let $\theta \in \mathbb{L}_I$. The Bayesian clustering model is then defined hierarchically as,

$$X \mid \theta \sim P_\theta, \quad \theta \mid I \sim \Pi_I, \quad I \sim \lambda_I$$

which induces a mixture prior on θ given by,

$$\theta \sim \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I. \quad (2.4)$$

Lemma 2.3. Let the data $X \sim P_\theta$ be according to (2.1), with $\theta \in \mathbb{L}_I$, where $I \in \mathcal{I}_k$ is unknown to the observer. Then, the marginal distribution for X , the posterior for I (given the data X) and, the posterior for θ (given the data X) are given by,

$$P_X = \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X, I_l, i}(x_i), \quad \Pi(I|X) = \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X, I_l, i}(x_i), \quad (2.5)$$

$$\Pi(\theta|X) = \frac{1}{P_X} \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i}(\theta_i) \cdot P_{\theta_i}(x_i), \quad (2.6)$$

respectively.

Proof of Lemma 2.3. Consider the Bayesian Clustering Model as given by Definition 2.2 and $X \sim P_\theta$ according to (2.1). Under the mixture assumptions on θ specified in (2.4), the marginal distribution of X is,

$$P_X = \sum_{I \in \mathcal{I}_k} \lambda_I P_{X, I} = \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{i \in [n]} P_{X, I, i}(x_i) = \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X, I_l, i}(x_i).$$

Similarly, the posterior for I (given X),

$$\Pi(I|X) = \frac{\lambda_I P_{X,I}}{P_X} = \frac{\lambda_I}{P_X} \bigotimes_{i \in [n]} P_{X,I,i}(x_i) = \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i).$$

Utilizing the previously derived expression for the posterior of I (given X), the posterior of θ (given X) is,

$$\begin{aligned} \Pi(\theta|X) &= \sum_{I \in \mathcal{I}_k} \Pi_I(\theta|X) \Pi(I|X) \\ &= \sum_{I \in \mathcal{I}_k} \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l,i}(\theta_i|X_i) \times \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i) \\ &= \frac{1}{P_X} \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l,i}(\theta_i|X_i) \times P_{X,I_l,i}(x_i). \end{aligned}$$

Finally, recalling that $\Pi_{I_l,i}(\theta_i|X_i) = \frac{\Pi_{I_l,i}(\theta_i) \cdot P_{\theta_i}(X_i)}{P_{X,I_l,i}(X_i)}$, for $i \in [n], l \in [k]$, we arrive at the required expression. $\square$

The problem of clustering can be reduced to the estimation of I_0 for the “true” $\theta_0 \in \mathbb{L}_{I_0}$. Consequently, the posterior distribution $\Pi(I | X)$ plays a central role in this study. In the subsequent sections, we extend the analysis from the case of 2-group clustering to the general case k -groups clustering setting, introducing the *product-form prior* and *cardinality-based prior*, which differ in the generation of the mixture weights λ_I ’s.

For notational simplicity, we denote the marginal distributions $P_{X,I_l,i}(\cdot)$ by $P_{X,l,i}(\cdot)$. Similarly, the prior distribution on θ , originally written as $\Pi_{I_l,i}$, will be abbreviated as $\Pi_{l,i}$. Correspondingly, the posterior distribution $\Pi_{I_l,i}(\theta | X)$ is denoted by $\Pi_{l,i}(\theta | X)$.

2.4 Binary Clustering

Similar to the sparsity setting, as given by (1.6), where two clusters —*on* and *off*—are formed, working with a single cluster is sufficient to determine the entire partition. Without loss of generality, we choose I_1 to define the partition I , such that $I = (I_1, I_1^c)$

and we proceed by assigning weights to the individual points in cluster I_1 .

2.4.1 Prior in Product-form

Definition 2.4. Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ follow the model (2.1). For a positive sequence $\lambda = (\lambda_i, i \in [n])$, define the *product-form* prior on the partition space $\mathcal{I}_2$ as follows,

$$I \sim \lambda_I^{prod}, \text{ where } \lambda_I^{prod} := C_\lambda \prod_{i \in I_1} \lambda_i \quad (2.7)$$

for some normalizing constant $C_\lambda \in \mathbb{R}_{>0}$.

Proposition 2.5 (Computation of C_λ). Assume the product-form prior on $\mathcal{I}_2$ as given by (2.7) and $I = (I_1, I_1^c)$, where $\lambda = (\lambda_i, i \in [n])$ is a positive sequence. It must hold that, $C_\lambda = \prod_{i=1}^n (1 + \lambda_i)^{-1}$.

Proof of Proposition 2.5. Suppose a prior λ on $\mathcal{I}_2$ expressed by, $\lambda_I = C_\lambda \prod_{i \in I_1} \lambda_i$, where $I = (I_1, I_1^c) \in \mathcal{I}_2$. Given the distribution assumptions of λ , it must hold that, $\sum_{I \in \mathcal{I}_2} \lambda_I = 1$. Combining the two expressions results in,

$$\sum_{I \in \mathcal{I}_2} \lambda_I = \sum_{I \in \mathcal{I}_2} C_\lambda \prod_{i \in I_1} \lambda_i = \sum_{I \in \mathcal{I}_2} C_\lambda \prod_{i \in I_1} \lambda_i \prod_{j \in I_1^c} 1.$$

The following elementary identity is true for any a_i and b_i , $\forall i \in [n]$,

$$\prod_{i \in [n]} (a_i + b_i) = \sum_{I \in \mathcal{I}_2} \left(\prod_{i \in I_1} a_i \right) \left(\prod_{j \in I_1^c} b_j \right). \quad (2.8)$$

By applying (2.8) for $a_i = \lambda_i$ and $b_i = 1$, $\forall i \in [n]$, we arrive at the expression, $\sum_{I \in \mathcal{I}_2} \lambda_I = C_\lambda \prod_{i \in [n]} (\lambda_i + 1)$. Thus, by requiring $C_\lambda = \prod_{i \in [n]} (\lambda_i + 1)^{-1}$, the prior distribution assumption of λ_I holds. Consequently,

$$\lambda_I = C_\lambda \prod_{i \in I_1} \lambda_i, \quad C_\lambda = \prod_{i \in [n]} (\lambda_i + 1)^{-1}.$$

□

It is trivial to show that, $\lambda_I \in [0, 1]$ for each $I \in \mathcal{I}_2$.

In this approach, the prior $I \sim \lambda_I$ is of the *product-form* based solely of individual weights of the “active” group I_1 . Therefore, the product-form prior is completely determined by the vector $(\lambda_i, i \in [n])$, where each λ_i corresponds to the membership of points i in the “active” cluster I_1 . Note that, unlike probabilities, weights λ_i are not restricted to the interval $[0, 1]$; rather, they can take any positive real value, i.e., $\lambda_i \in (0, \infty)$.

Theorem 2.6. *Let $X = (X_1, \dots, X_n) \sim P_\theta$ be according to the model (2.1), with $\theta \in \mathbb{L}_I$, where unknown to the observer $I = (I_1, I_1^c) \in \mathcal{I}_2$ and the prior Π , as be defined by (2.4) with a prior λ_I of the product form (according to (2.7) from Definition 2.4). Then, the marginal distribution for X and the posterior distributions for I and θ , respectively, can be presented as follows,*

$$\begin{aligned}\Pi &= \bigotimes_{j \in [n]} \left[\omega_j \Pi_{1,j} + (1 - \omega_j) \Pi_{2,j} \right], & P_X &= \bigotimes_{j \in [n]} \left[\omega_j P_{X,1,j}(x_j) + (1 - \omega_j) P_{X,2,j}(x_j) \right], \\ \Pi(I|X) &= \frac{1}{P_X} \bigotimes_{j \in I_1} \omega_j P_{X,1,j}(x_j) \times \bigotimes_{j \in I_1^c} (1 - \omega_j) P_{X,2,j}(x_j), \\ \Pi(\theta|X) &= \frac{1}{P_X} \bigotimes_{j \in [n]} \left[\omega_j \Pi_{1,j}(\theta_j) P_{\theta_j}(x_j) + (1 - \omega_j) \Pi_{2,j}(\theta_j) P_{\theta_j}(x_j) \right],\end{aligned}$$

where, $\omega_j = \frac{\lambda_j}{1 + \lambda_j}$, $j \in [n]$.

Proof of Theorem 2.6. By substituting the product-form prior $I \sim \lambda_I^{prod}$ into the expressions given by (2.5) in Lemma 2.3, we obtain,

$$\begin{aligned}\Pi &= \sum_{I \in \mathcal{I}_2} \lambda_I^{prod} \Pi_I = \sum_{I \in \mathcal{I}_2} C_\lambda \prod_{i \in I_1} \lambda_i \left[\bigotimes_{i \in I_1} \Pi_{1,i} \times \bigotimes_{j \in I_1^c} \Pi_{2,j} \right] \\ &= C_\lambda \sum_{I \in \mathcal{I}_2} \left(\bigotimes_{i \in I_1} \lambda_i \Pi_{1,i} \times \bigotimes_{j \in I_1^c} \Pi_{2,j} \right)\end{aligned}$$

By using the previously presented, elementary identity, given by (2.8) for $a_i = \Pi_{1,i}$ and $b_i = \Pi_{2,i}$, $\forall i \in [n]$, it follows that,

$$\Pi = \prod_{i \in [n]} (1 + \lambda_i)^{-1} \bigotimes_{i \in [n]} \left[\lambda_i \Pi_{1,i} + \Pi_{2,i} \right]$$

$$= \bigotimes_{i \in [n]} \left[\lambda_i (\lambda_i + 1)^{-1} \Pi_{1,i} + (1 + \lambda_i)^{-1} \Pi_{2,i} \right]$$

By setting $\omega_i = \frac{\lambda_i}{1+\lambda_i}$, for all $i \in [n]$, we arrive at the simplified expression $\Pi = \bigotimes_{i \in [n]} \left[\omega_i \Pi_{1,i} + (1 - \omega_i) \Pi_{2,i} \right]$. Similarly for the marginal distribution for X ,

$$\begin{aligned} P_X &= \sum_{I \in \mathcal{I}_2} \lambda_I P_{X,I}(x) = \sum_{I \in \mathcal{I}_2} \left(C_\lambda \Pi_{i \in I_1} \lambda_i \bigotimes_{i \in I_1} P_{X,1,i}(x_i) \times \bigotimes_{j \in I_1^c} P_{X,2,j}(x_j) \right) \\ &= C_\lambda \sum_{I \in \mathcal{I}_2} \left(\bigotimes_{i \in I_1} \lambda_i P_{X,1,i}(x_i) \times \bigotimes_{j \in I_1^c} P_{X,2,j}(x_j) \right) \\ &\stackrel{(2.8)}{=} C_\lambda \bigotimes_{i \in [n]} \left[\lambda_i P_{X,1,i}(x_i) + P_{X,2,i}(x_i) \right] \\ &= \bigotimes_{i \in [n]} \left[\omega_i P_{X,1,i}(x_i) + (1 - \omega_i) P_{X,2,i}(x_i) \right]. \end{aligned}$$

By applying the same reasoning, based on the assumption $I \sim \lambda_I^{prod}$, we obtain the posterior for I (given X) as,

$$\begin{aligned} \Pi(I|X) &= \frac{\lambda_I P_{X,I}}{P_X} = \frac{C_\lambda}{P_{X,I}} \prod_{i \in I_1} \lambda_i P_{X,i} \\ &= \frac{C_\lambda}{P_X} \prod_{i \in I_1} \lambda_i \bigotimes_{i \in I_1} P_{X,1,i}(x_i) \times \bigotimes_{j \in I_1^c} P_{X,2,j}(x_j) \\ &= \frac{C_\lambda}{P_X} \bigotimes_{i \in I_1} \lambda_i P_{X,1,i}(x_i) \times \bigotimes_{j \in I_1^c} P_{X,2,j}(x_j) \\ &= \frac{1}{P_X} \bigotimes_{i \in I_1} \omega_i P_{X,1,i}(x_i) \times \bigotimes_{j \in I_1^c} (1 - \omega_j) P_{X,2,j}(x_j). \end{aligned}$$

Finally, by combining the previously derived expression for $\Pi(I|X)$ together with the expression $\Pi_{I,l,i}(\theta_i|X_i) = \Pi_{I,l,i}(\theta_i) \cdot P_{\theta_i}(X_i) / P_{X,I,l,i}(X_i)$ for $i \in [n], l \in [k]$, we obtain the

desired for the posterior expression for θ (given X),

$$\begin{aligned}
\Pi(\theta|X) &= \sum_{I \in \mathcal{I}_2} \Pi_I(\theta|X) \Pi(I|X) \\
&= \frac{C_\lambda}{P_X} \sum_{I \in \mathcal{I}_2} \left(\bigotimes_{i \in I_1} \lambda_i \Pi_{1,i}(\theta_i) P_{\theta_i}(x_i) \times \bigotimes_{j \in I_1^c} \Pi_{2,j}(\theta_j) P_{\theta_j}(x_j) \right) \\
&\stackrel{(2.8)}{=} \frac{1}{P_X} \bigotimes_{i \in [n]} [\omega_i \Pi_{1,i}(\theta_i) P_{\theta_i}(x_i) + (1 - \omega_i) \Pi_{2,i}(\theta_i) P_{\theta_i}(x_i)].
\end{aligned}$$

□

Remark 1. Notice that, ω_j can be interpreted as the prior probability that point j belongs to cluster I_1 and $1 - \omega_j$ as the prior probability that point j belongs to I_1^c . Assume $\omega_j = \mathbb{P}(j \in I_1) = p$, $i \in [n]$. Then, the prior probability of a clustering $I = (I_1, I_1^c)$ is given by $p^{|I_1|}(1-p)^{(n-|I_1|)}$, which implies that $|I_1| \sim \text{Binom}(n, p)$, providing with a connection between the product-form and cardinality-based prior.

2.4.2 Cardinality-based prior

In this case, we deviate from the product-form assumption and introduce a prior on the cardinality of group I_1 .

Definition 2.7. Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ follow the model (2.1), with $\theta \in \mathbb{L}_I$ where, $I = (I_1, I_1^c) \in \mathcal{I}_2$. The *cardinality-based* prior λ^{card} on $\mathcal{I}_2$, is defined as follows,

$$I \sim \lambda_I^{card} \quad \text{where,} \quad \lambda_I^{card} := g(|I_1|) \binom{n}{|I_1|}^{-1}, \quad |I_1| \in [n]_0 \quad (2.9)$$

where, $g(\cdot)$ is a probability distribution on $[n]_0$, meaning that, $\sum_{m \in [n]_0} g(m) = 1$ and $g(m) \in [0, 1]$.

Indeed, the cardinality-based prior $(\lambda_I, I \in \mathcal{I}_2)$ defines a probability distribution over $\mathcal{I}_2$, as shown in Appendix A.1.

The following Lemma introduces the so called “polynomial trick” which serves as basis for posterior pointwise computations. For the 2-group prior, this idea was first exploited in [9] in the sparsity context and in Belitser [3] for the clustering context.

Lemma 2.8 (Polynomial trick). For vectors $\alpha = (\alpha_i, i \in [n]) \in \mathbb{R}^n, \beta = (\beta_i, i \in [n]), \beta \in \mathbb{R}^n$, define the polynomial,

$$\Phi_{\alpha, \beta}(Z) = \prod_{i=1}^n (\alpha_i + \beta_i Z).$$

It follows that the coefficient of Z^m of $\Phi_{\alpha, \beta}(Z)$ is given by,

$$A_m(\alpha, \beta) = \sum_{|S|=m} \prod_{i \notin S} \alpha_i \prod_{i \in S} \beta_i$$

where $I = (S, S^c) \in \mathcal{I}_2$.

Proof of Lemma 2.8. For $\alpha = (\alpha_i, i \in [n]) \in \mathbb{R}^n$ and $\beta = (\beta_i, i \in [n]) \in \mathbb{R}^n$, define the polynomial $\Phi_{\alpha, \beta}(Z) \in \mathbb{R}[Z]$ as

$$\Phi_{\alpha, \beta}(Z) = \prod_{i=1}^n (\alpha_i + \beta_i Z) = A_0 + A_1 Z + \dots A_n Z^n,$$

where $A_m = A_m(\alpha, \beta)$. For some set $S \subseteq [n]$, with exactly m indices, $|S| = m$, we have,

$$\prod_{i \in S} (\beta_i \cdot Z) \prod_{i \in S^c} \alpha_i = \left(\prod_{i \in S} \beta_i \prod_{i \in S^c} \alpha_i \right) Z^{|S|}.$$

Considering all subsets $S \subseteq [n]$, we obtain the m -order coefficient,

$$A_m(\alpha, \beta) = \sum_{|S|=m} \left(\prod_{i \in S} \beta_i \prod_{i \in S^c} \alpha_i \right).$$

where each $I = (S, S^c)$ defines a partition of $[n]$ into 2-groups, $I \in \mathcal{I}_2$. □

Theorem 2.9. Let the data $X = (X_1, \dots, X_n) \sim P_\theta$ be according to model (2.1), with $\theta \in \mathbb{L}_I$, where unknown to the observer $I = (I_1, I_1^c) \in \mathcal{I}_2$ and the prior Π be defined by (2.4) with a cardinality-based prior (according to (2.9) from Definition 2.7), $I \sim \lambda_I^{\text{card}}$. Denote $h(m) := g(m) \binom{n}{m}^{-1}$, where $m = |I_1|$. Then, the marginal distribution for X and the posterior distributions for I and θ , respectively, are represented as follows,

$$\Pi = \sum_{m=0}^n h(m) A_m(\Pi_1, \Pi_2), \quad P_X = \sum_{m=0}^n h(m) A_m(P_{X,1}, P_{X,2}),$$

$$\Pi(I|X) = h(|I_1|) \frac{P_{X,I}}{P_X} \quad \text{and,} \quad \Pi(\theta|X) = \frac{\sum_{m=0}^n h(m) A_m(\rho_1, \rho_2)}{P_X},$$

where, $\Pi_l = (\Pi_{l,i}, i \in [n])$, $P_{X,l} = (P_{X,l,i}(x_i), i \in [n])$, $\rho_l = (\Pi_{l,i}(\theta_i)P_{\theta_i}(x_i), i \in [n])$ for $l = 1, 2$ and $A_m(\cdot)$ the m^{th} coefficient of the polynomial as given in Lemma 2.8.

Proof of Theorem 2.9. Following Definition 2.7 for the cardinality-based prior, the prior on θ (Π), the marginal distribution of X , and the posterior distribution of θ can be expressed as follows,

$$\Pi = \sum_{I \in \mathcal{I}_2} \lambda_I \Pi_I = \sum_{m=0}^n \sum_{I: |I_1|=m} h(m) \left(\bigotimes_{i \in I_1} \Pi_{1,i} \times \bigotimes_{j \in I_1^c} \Pi_{2,j} \right), \quad (2.10)$$

$$P_X = \sum_{I \in \mathcal{I}_2} \lambda_I P_{X,I} = \sum_{m=0}^n \sum_{I: |I_1|=m} \left(h(m) \bigotimes_{i \in I_1} P_{X,1,i}(x_i) \times \bigotimes_{j \in I_1^c} P_{X,2,j}(x_j) \right), \quad (2.11)$$

$$\Pi(\theta|X) = \frac{1}{P_X} \sum_{m=0}^n \sum_{I: |I_1|=m} h(m) \bigotimes_{i \in I_1} \Pi_{1,i}(\theta_i) P_{\theta_i}(x_i) \times \bigotimes_{j \in I_1^c} \Pi_{2,j}(\theta_j) P_{\theta_j}(x_j). \quad (2.12)$$

Let the vectors, $\Pi_1 = (\Pi_{1,i}, i \in [n])$ and $\Pi_2 = (\Pi_{2,i}, i \in [n])$. By applying the polynomial trick (Lemma 2.8) on the expression (2.10) for Π , it follows that,

$$\Pi = \sum_{m=0}^n h(m) \sum_{I: |I_1|=m} \left(\bigotimes_{i \in I_1} \Pi_{1,i} \times \bigotimes_{j \in I_1^c} \Pi_{2,j} \right) = \sum_{m=0}^n h(m) A_m(\Pi_1, \Pi_2).$$

Similarly, suppose the pair $P_{X,1} = (P_{X,1,i}(x_i), i \in [n])$, $P_{X,2} = (P_{X,2,i}(x_i), i \in [n])$ for the marginal distribution for X and the pair $(\rho_1 = (\rho_1(x_i), i \in [n]), \rho_2 = (\rho_2(x_i), i \in [n]))$, where $\rho_l(x_i) = \Pi_{l,i}(\theta_i)P_{\theta_i}(x_i)$ for $l = 1, 2$, for the posterior for θ (given X). By applying the polynomial trick for the expressions (2.11), (2.12), we arrive at the required result. $\square$

2.5 k -Groups Clustering

In this chapter, we extend the Clustering analysis from 2-groups to the general case of k -groups. For the cardinality-based prior case, this extension is achieved by adapting the polynomial trick from the 2-group setting into a multivariate polynomial trick that naturally handles multiple clusters.

2.5.1 Prior in Product-form

Similarly to the multi-level sparsity for unknown cluster values (1.7), we extend into general clustering into k -groups. Without loss of generality and since $k-1$ groups are sufficient to determine the entire partition,

$$I = (I_1, I_2, \dots, I_{k-1}, I_k), \quad I_k = [n] \setminus \bigcup_{l=1}^{k-1} I_l$$

we continue by assigning weights to $k-1$ groups.

Definition 2.10. Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ follow the model (2.1). For a positive matrix $\Lambda = (\lambda_{l,j}, j \in [n], l \in [k-1])$, define the *product-form* prior on $\mathcal{I}_k$ as follows,

$$I \sim \lambda_I^{prod}, \text{ where } \lambda_I^{prod} = C_\lambda \prod_{l \in [k-1]} \prod_{i \in I_l} \lambda_{l,i} \quad (2.13)$$

with $I = (I_1, \dots, I_k) \in \mathcal{I}_k$, for some normalizing constant $C_\lambda \in \mathbb{R}_{>0}$.

Proposition 2.11 (Computation of C_λ). Assume the product-form prior on $\mathcal{I}_k$ as given by (2.10) and $I = (I_1, \dots, I_k) \in \mathcal{I}_k$ where $\Lambda = (\lambda_{l,i}, i \in [n], l \in [k-1])$ is a positive matrix. It must hold that, $C_\lambda = \prod_{i=1}^n (\lambda_{1,i} + \dots + \lambda_{k-1,i} + 1)^{-1}$.

Proof of Proposition 2.11. Since λ is a prior distribution on $\mathcal{I}_k$ such that, $\lambda_I = C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i}$ it must hold that $\sum_{I \in \mathcal{I}_k} \lambda_I = 1$. Combining the two expressions results in,

$$\begin{aligned} \sum_{I \in \mathcal{I}_k} \lambda_I &= \sum_{I \in \mathcal{I}_k} C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i} \\ &= \sum_{I \in \mathcal{I}_k} C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i} \prod_{j \in I_k} 1. \end{aligned}$$

By using the following elementary identity,

$$\prod_{i \in [n]} (a_{1,i} + \dots + a_{k,i}) = \sum_{I \in \mathcal{I}_k} \left(\prod_{i \in I_1} a_{1,i} \right) \cdots \left(\prod_{i \in I_k} a_{k,i} \right), \quad (2.14)$$

for any $\alpha_{l,i}, i \in [n], l \in [k-1]$, we arrive at the expression,

$$\sum_{I \in \mathcal{I}_k} \lambda_I = C_\lambda \sum_{I \in \mathcal{I}_k} \left(\prod_{i \in I_1} \lambda_{1,i} \right) \cdots \left(\prod_{i \in I_{k-1}} \lambda_{k-1,i} \right) \left(\prod_{i \in I_k} 1 \right).$$

Thus, by requiring $C_\lambda := \prod_{i=1}^n (\lambda_{1,i} + \cdots + \lambda_{k-1,i} + 1)^{-1}$, the prior assumption of λ_I holds. Consequently,

$$\lambda_I^{prod} = C_\lambda \prod_{l \in [k]} \prod_{i \in [n]} \lambda_{l,i}, \text{ where, } C_\lambda := \prod_{i=1}^n (\lambda_{1,i} + \cdots + \lambda_{k-1,i} + 1)^{-1}.$$

□

Assuming the product-form prior for arbitrary k -groups on θ , the expressions used in the Bayesian analysis are adapted to k -groups clustering in the following Theorem.

Theorem 2.12. *Let the data $X = (X_1, \dots, X_n) \sim P_\theta$ be according to model (2.1), with $\theta \in \mathbb{L}_I$, where unknown partition to the observer $I = (I_1, I_2, \dots, I_k) \in \mathcal{I}_k$ and the prior Π defined by (2.4) with a product-form prior (according to (2.13) from Definition 2.10), $I \sim \lambda_I^{prod}$. Then, the marginal distribution for X and the posterior distributions for I and θ , respectively, are represented as follows,*

$$P_X = \bigotimes_{i \in [n]} \left[\sum_{l=1}^k \omega_{l,i} P_{X,l,i}(x_i) \right], \quad \Pi = \bigotimes_{i \in [n]} \left[\sum_{l=1}^k \omega_{l,i} \Pi_{l,i}(x_i) \right],$$

$$\Pi(I|X) = \frac{1}{P_X} \bigotimes_{l=1}^k \bigotimes_{i \in I_l} \omega_{l,i} P_{X,l,i}(x_i), \quad \Pi(\theta|X) = \frac{1}{P_X} \bigotimes_{i \in [n]} \left[\sum_{l=1}^k \omega_{l,i} \Pi_{l,i}(\theta_i) \cdot P_{\theta_i}(X_i) \right],$$

where,

$$\omega_{l,j} = \frac{\lambda_{l,j}}{\sum_{s=1}^{k-1} \lambda_{s,j} + 1}, \quad l \in [k-1] \quad \text{and,} \quad \omega_{k,j} = 1 - \sum_{s=1}^{k-1} \omega_{s,j}, \quad j \in [n].$$

Proof of Theorem 2.12. Assume $X = (X_1, \dots, X_n) \sim P_\theta$, with $\theta \in \mathbb{L}_I$ and $I \in \mathcal{I}_2$ and that the priors on θ and I are given by, $\theta \sim \Pi$ and $I \sim \lambda_I^{prod}$, respectively. We begin by

computing the marginal distribution of X ,

$$\begin{aligned}
P_X &= \sum_{I \in \mathcal{I}_k} \left(C_\lambda \cdot \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,l,i}(x_i) \right) \\
&= \sum_{I \in \mathcal{I}_k} \left(C_\lambda \prod_{l \in [k-1]} \prod_{i \in I_l} \lambda_{l,i} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,l,i}(x_i) \right) \\
&= C_\lambda \sum_{I \in \mathcal{I}_k} \left(\bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l,i} P_{X,l,i}(x_i) \times \bigotimes_{j \in I_k} P_{X,k,j}(x_j) \right) \\
&\stackrel{(2.14)}{=} C_\lambda \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \lambda_{l,i} P_{X,l,i}(x_i) + P_{X,k,i}(x_i) \right].
\end{aligned}$$

By setting, $\omega_{l,i} = \frac{\lambda_{l,i}}{\sum_{s=1}^{k-1} \lambda_{s,i} + 1}$ for $l \in [k-1]$ and $\omega_{k,i} = 1 - \sum_{s=1}^{k-1} \omega_{s,i}$ for $i \in [n]$, we arrive the required expression. Similarly, for the prior on θ we assume,

$$\begin{aligned}
\Pi &= C_\lambda \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{l,i} \\
&= C_\lambda \sum_{I \in \mathcal{I}_k} \left(\bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l,i} \Pi_{l,i}(x_i) \times \bigotimes_{j \in I_k} \Pi_{I_k,j}(x_j) \right) \\
&\stackrel{(2.14)}{=} C_\lambda \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \lambda_{l,i} \Pi_{l,i}(x_i) + \Pi_{I_k,i}(x_i) \right] = \bigotimes_{i \in [n]} \left[\sum_{l \in [k]} \omega_{l,i} \Pi_{l,i}(x_i) \right].
\end{aligned}$$

Using the previously derived expression for P_X , the posterior for I given the data X is then expressed as,

$$\begin{aligned}
\Pi(I|X) &= \frac{\lambda_I P_{X,I}}{P_X} = \frac{C_\lambda}{P_X} \prod_{l \in [k-1]} \prod_{i \in I_l} \lambda_{l,i} P_{X,l,i} \\
&= \frac{C_\lambda}{P_X} \bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l,i} P_{X,l,i}(x_i) \times \bigotimes_{j \in I_k} P_{X,k,j}(x_j) \\
&= \frac{1}{P_X} \bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \omega_{l,i} P_{X,l,i}(x_i).
\end{aligned}$$

Analogously, using the previously derived expression for Π , the posterior for θ given

the data X is expressed as

$$\begin{aligned}
\Pi(\theta|X) &= \sum_{I \in \mathcal{I}_k} \Pi_I(\theta|X) \Pi(I|X) = \sum_{I \in \mathcal{I}_k} \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{l,i}(\theta_i) \cdot P_{\theta_i}(X_i) \\
&= \frac{C_\lambda}{P_X} \sum_{I \in \mathcal{I}_k} \left[\bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l,i} \Pi_{l,i}(\theta_i) \cdot P_{\theta_i}(X_i) \times \bigotimes_{j \in I_k} \Pi_{I_k,j}(\theta_j) \cdot P_{\theta_j}(X_j) \right] \\
&= \frac{1}{P_X} \bigotimes_{i \in [n]} \left[\sum_{l \in [k]} \omega_{l,i} \Pi_{l,i}(\theta_i) \cdot P_{\theta_i}(X_i) \right]
\end{aligned}$$

which concludes the proof. $\square$

2.5.2 Cardinality-based prior

Definition 2.13. Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ follow the model (2.1) and $I = (I_1, \dots, I_k) \in \mathcal{I}_k$. The *cardinality-based k -groups prior* on $\mathcal{I}_k$ is defined as follows,

$$I \sim \lambda_I^{card}, \quad \text{where} \quad \lambda_I^{card} = c_{\mathbf{m}} g(\mathbf{m}), \quad c_{\mathbf{m}} = \left(\binom{n}{m_1, m_2, \dots, m_k} \right)^{-1}, \quad (2.15)$$

with $\mathbf{m} = (m_1, \dots, m_k)$ the multi-array expressing the cardinalities of groups $I_1, \dots, I_k$ and $g(\cdot)$ the probability distribution on $\mathcal{M}$, such that, $\sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) = 1$ with,

$$\mathcal{M} = \{\mathbf{m} = (m_1, \dots, m_k) \in \mathbb{N}_0^k : m_1 = |I_1|, \dots, m_k = |I_k|\}.$$

Similarly to the 2-group case, the prior $(\lambda_I^{card}, I \in \mathcal{I}_k)$ determines a distribution on $\mathcal{I}_k$, as shown in Appendix A.1 (as it is done in Belitser [3]). Moreover, the polynomial trick for 2-groups is generalized in the following multivariate polynomial.

Lemma 2.14 (Multivariate polynomial Trick). *For a matrix, $\alpha = (\alpha_{li}, l \in [k], i \in [n])$, define the multivariate polynomial $\Phi_\alpha(Z)$ as*

$$\Phi_\alpha(Z) = \prod_{i=1}^n (\alpha_{1i} Z_1 + \alpha_{2i} Z_2 + \dots + \alpha_{ki} Z_k)$$

where, $Z = (Z_1, \dots, Z_k) \in \mathbb{R}^k$ are the polynomial variables. Then, the coefficients of the

multivariate polynomial are expressed by,

$$A_{\mathbf{m}}(a_1, \dots, a_k) = \sum_{|S|=\mathbf{m}} \prod_{i \in S_1} \alpha_{1i} \prod_{i \in S_2} \alpha_{2i} \cdots \prod_{i \in S_k} \alpha_{ki}, \quad (2.16)$$

where, $\alpha_1, \dots, \alpha_k$ are the row-vectors of matrix α , $\mathbf{m} = (m_1, m_2, \dots, m_k)$ is the mult-array representing the cardinalities of groups $S_1, \dots, S_k$ and $S = (S_1, \dots, S_k) \in \mathcal{I}_k$.

Proof of Lemma 2.14, Multivariate Polynomial Trick . For $\alpha = (a_{li}, i \in [n], l \in [k]) \in \mathbb{R}^n$, define the multivariate polynomial $\Phi_{\alpha, \beta}(Z) \in \mathbb{R}[Z]$ as

$$\Phi_{\alpha}(Z) = \prod_{i=1}^n (a_{1,i}Z_1 + a_{2,i}Z_2 + \dots a_{k,i}Z_k)$$

Its $\mathbf{m}^{th}$ -order coefficient, $A_{\mathbf{m}}$ is the coefficient of the monomial, $Z_1^{m_1} Z_2^{m_2} \cdots Z_k^{m_k}$. Using the expansion of the polynomial, choose exactly m_1 indices, $i \in S_1 \subseteq [n]$, then choose m_2 indices, $i \in S_2 \subseteq [n - m_1]$, repeat till m_{k-1} , $i \in S_{k-1} \subseteq [n - (m_1 + \dots + m_{k-2})]$. It follows that, $|S| = \mathbf{m}$ and

$$\prod_{i \in S_1} (a_{1,i} \cdot 1) \prod_{i \in S_2} (a_{2,i} \cdot Z_2) \cdots \prod_{i \in S_k} (a_{k,i} \cdot Z_k) = \left(\prod_{i \in S_1} a_{1,i} \right) \cdots \left(\prod_{i \in S_2} a_{2,i} \right) \left(\prod_{i \in S_k} a_{k,i} \right) Z_2^{m_1} \cdots Z_k^{m_k},$$

where, for simplicity, $Z_1 \equiv 1$.

Considering all subsets $S_1, \dots, S_k \subseteq [n]$, so that, $|S| = \mathbf{m}$, it holds that $S = (S_1, \dots, S_k)$ is a partition of $[n]$ indices into k -groups, $S \in \mathcal{I}_k$,

$$A_{\mathbf{m}}(a_1, \dots, a_k) = \sum_{|S|=\mathbf{m}} \left(\prod_{i \in S_1} a_{1,i} \prod_{i \in S_2} a_{2,i} \cdots \prod_{i \in S_k} a_{k,i} \right).$$

□

Theorem 2.15. *Let the data $X = (X_1, \dots, X_n) \sim P_{\theta}$ be according to model (2.1), with $\theta \in \mathbb{L}_I$, where unknown to the observer $I = (I_1, \dots, I_k) \in \mathcal{I}_k$ and the prior Π defined by (2.4) with a cardinality-based prior for k -groups (according to (2.15) from Definition 2.13), $I \sim \lambda_I^{card}$. Set $h(\mathbf{m}) = g(\mathbf{m}) \binom{n}{m_1, \dots, m_k}^{-1}$, $\mathbf{m} \in \mathcal{M}$. Then, the marginal distribution for X and*

the posterior distributions for I and θ , respectively, are represented as follows,

$$\begin{aligned}\Pi &= \sum_{\mathbf{m} \in \mathcal{M}} h(\mathbf{m}) A_{\mathbf{m}}(\Pi_1, \dots, \Pi_k), \quad P_X = \sum_{\mathbf{m} \in \mathcal{M}} h(\mathbf{m}) A_{\mathbf{m}}(P_{X,1}, \dots, P_{X,k}) \\ \Pi(I|X) &= g(|I_1|, \dots, |I_k|) \binom{n}{|I_1|, \dots, |I_k|}^{-1} \frac{P_{X,I}}{P_X}, \quad \Pi(\theta|X) = \frac{1}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} \sum_{I: |I|=\mathbf{m}} h(\mathbf{m}) A_{\mathbf{m}}(\rho_1, \dots, \rho_k),\end{aligned}$$

where for $l \in [k]$, Π_l , $P_{X,l}$, ρ_l are the row-vectors of matrices $\boldsymbol{\pi} = (\Pi_{l,i}, i \in [n], l \in [k])$, $\mathbf{p} = (P_{X,l,i}(x_i), i \in [n], l \in [k])$, $\boldsymbol{\rho} = (\Pi_{l,i}(\theta_i) P_{\theta_i}(x_i), i \in [n], l \in [k])$, respectively and $A_{\mathbf{m}}(\cdot)$ as defined by (2.20) represents the coefficient of monomial of order $\mathbf{m} = (m_1, \dots, m_k)$ of the respective variables $Z_1, \dots, Z_k$ of the multi-variate polynomial, as given in Lemma 2.14.

Proof of Theorem 2.15. Given the data $X \sim P_{\theta}$, with $\theta \in \mathbb{L}_I$ and the prior assumptions, $I \sim \lambda_I^{card}$, with $I = (I_1, \dots, I_k) \in \mathcal{I}_k$, the marginal distribution of X , the prior on θ and the posterior for I and θ , transform, respectively, into,

$$P_X = \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,l,i}(x_i) = \sum_{\mathbf{m} \in \mathcal{M}} \sum_{I: |I|=\mathbf{m}} \left(h(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,l,i}(x_i) \right), \quad (2.17)$$

$$\Pi = \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I = \sum_{\mathbf{m} \in \mathcal{M}} \sum_{I: |I|=\mathbf{m}} \left(h(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{l,i} \right), \quad (2.18)$$

$$\Pi(\theta|X) = \sum_{I \in \mathcal{I}_k} \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{l,i}(\theta_i) \cdot P_{\theta_i}(x_i). \quad (2.19)$$

The expression of $\Pi(I|X)$ follows directly from the Definition 2.13 of $I \sim \lambda_I^{card}$. By applying the multivariate polynomial trick to P_X , as given in expression (2.17) using the matrix $\mathbf{p} = (P_{X,l,i}(x_i), i \in [n], l \in [k])$, we arrive at the required expression,

$$P_X = \sum_{\mathbf{m} \in \mathcal{M}} h(\mathbf{m}) A_{\mathbf{m}}(P_{X,I_1}, P_{X,I_2}, \dots, P_{X,I_k}).$$

Similarly, for expression (2.18), consider the matrix $\boldsymbol{\pi} = (\Pi_{l,i}, i \in [n], l \in [k])$ and lastly, for expression (2.19), consider the matrix $\boldsymbol{\rho} = (\Pi_{l,i}(\theta_i) P_{\theta_i}(x_i), i \in [n], l \in [k])$. By applying the multivariate polynomial trick, respectively, we arrive at the required expressions. $\square$

2.6 Prediction

The posterior $\Pi(I|X)$ is used for inference on the partition of θ into k groups of coordinates. To quantify the posterior the *Maximum A-Posteriori* (MAP) estimator is proposed,

$$\hat{I}_{MAP} = \arg \max_{I \in \mathcal{I}_k} \Pi(I|X) = \arg \max_{I \in \mathcal{I}_k} \lambda_I P_{X,I}.$$

An approach to derive such an estimator is the **brute-force** setting of parsing the partition space, as demonstrated in Algorithm 1 for the product-form prior and Algorithm 2 for the cardinality based prior.

A major drawback of this method is that the size of the partition space grows exponentially (k^n) with the number of observations n , making such algorithms computationally infeasible for moderate to large datasets. Thus, an alternative approach should be introduced.

1. Product-form prior: Partition Space

Input: Data $x_1, \dots, x_n \sim P_\theta$, matrix $\Lambda = (\lambda_{li}) \in \mathbb{R}^{(k-1) \times N}$, prior means $\mu_1, \dots, \mu_k$ and prior $\theta \sim \Pi_\theta$

Output: Partition Estimator $\hat{I}$

// MAP-Estimator

- 1: Generate the partition space $\mathcal{I}_k$
 - 2: **for** each $l \in [k]$ **do**
 - 3: **for** each $i \in [n]$ **do**
 - 4: Compute $a_l \leftarrow (P_{X,l,i}(x_i))_{l,i}$
 - 5: **end for**
 - 6: **end for**
 - 7: Compute normalizing constants, C_λ, P_X
 - 8: Compute posterior mixture weights $\omega_{li} = \frac{\lambda_{li}}{\sum_{s=1}^{k-1} \lambda_{si} + 1}$, and $\omega_{ki} = 1 - \sum_{s=1}^{k-1} \omega_{si}$
 - 9: **for** each $I \in \mathcal{I}_k$ **do**
 - 10: Compute $\lambda_I = C_\lambda \cdot \prod_{l \in [k]} \prod_{i \in I_l} \lambda_{li}$
 - 11: Compute marginal likelihood $P_{X,I}(x) = \prod_{l \in [k]} \prod_{i \in I_l} P_{X,l,i}(x_i)$
 - 12: Store product $\Pi(I|X) = \frac{\lambda_I \cdot P_{X,I}(x)}{P_X}$
 - 13: **end for**
 - 14: **return** Partition $\hat{I} = \arg \max_I \Pi(I|x)$
-

Remark 2. Due to floating-point precision limits, repeated multiplication of the prior weights λ can cause the values to vanish. The constants P_X and C_λ counteract this by

stabilizing the computation.

2. Cardinality-Based Prior: Partition Space

Input: Data $x_1, \dots, x_n \sim P_\theta$, prior means $\mu_1, \dots, \mu_k$, prior $\theta \sim \Pi_\theta$ and cardinality weights $g_1, \dots, g_s$ with $\sum g = 1, (\mathbf{m} \rightarrow g(\mathbf{m}))$

Output: Partition estimate $\hat{I}$

// MAP-Estimator

```

1: Generate partition space  $\mathcal{I}_k$ 
2: for each  $l \in [k]$  do
3:   for each  $i \in [n]$  do
4:     Compute the marginal likelihood  $a_l \leftarrow (P_{X,l,i}(x_i))_{l,i}$ 
5:   end for
6: end for
7: Compute polynomial  $Z \rightarrow P[a_1, \dots, a_k](Z)$  // Lemma 2.14
8: Extract powers of monomials of P to get cardinalities space  $\mathcal{M}$ 
9: for each cardinality  $\mathbf{m} \in \mathcal{M}$  do
10:   Assign prior weight  $g_{\mathbf{m}}$  to  $\mathbf{m}$ 
11:   Extract coefficients  $A_{\mathbf{m}}$ 
12: end for
13: for partition  $I$  in  $\mathcal{I}_k$  do
14:   Compute cardinalities  $\mathbf{m} \leftarrow (|I_1|, \dots, |I_k|)$ 
15:   Compute marginal likelihood  $P_{X,I}(x) = \prod_{l \in [k]} \prod_{i \in I_l} P_{X,l,i}(x_i)$ 
16:   Compute  $\lambda_I = \binom{n}{\mathbf{m}}^{-1} g(\mathbf{m})$ 
17: end for
18: Compute  $P_X = \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}}(P_{X,1}(x), P_{X,2}(x), \dots, P_{X,k}(x))$ 
19: return Partition  $\hat{I} = \arg \max_I \Pi(I|x)$ 

```

2.6.1 Point-wise Posterior

In the clustering setting for unknown partition where a product-form prior is assumed on I , it holds that the posterior $\Pi(I|X) = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi(i \in I_l|X)$ or, alternatively, $\Pi(I|X) = \bigotimes_{i \in [n]} \Pi(c_i|X)$ and the MAP-estimator simplifies to,

$$\hat{I}_{MAP} = \arg \max_{I \in \mathcal{I}_k} \Pi(I|X) = \arg \max_{c_1, \dots, c_n} \prod_{i \in [n]} \Pi(c_i|X)$$

where $c_1, \dots, c_n$ is an equivalent expression of partition $I \in \mathcal{I}_k$.

Definition 2.16. Let $X = (X_1 \dots X_n) \in \mathbb{R}^n$, according to model (2.1), with $\theta \in \mathbb{L}_I$ and unknown $I = (I_1, \dots, I_k) \in \mathcal{I}_k$. We define the point-wise posterior by,

$$p_{l*,i} := \Pi(i \in I_{l*}|X) = \sum_{I: i \in I_{l*}} \Pi(I|X).$$

In case of the cardinality-based prior, the MAP estimate does *not* simplify to the component -wise posterior due to the combinatorial dependence of the cardinalities of partition I . However, the point-wise posterior can, still, be used independently to derive an estimator, namely, the Maximum Point-wise A-Posteriori Estimator, $\hat{I} = \hat{I}_{MPAP}$.

3. Product-form prior: point-wise posterior

Input: Data $x_1, \dots, x_n \sim P_\theta$, matrix $\Lambda = (\lambda_{li}) \in \mathbb{R}^{(k-1) \times N}$, prior means $\mu_1, \dots, \mu_k$ and prior $\theta \sim \Pi_\theta$

Output: Partition Estimator $\hat{I}$ // w.r.t. MPAP

- 1: Compute normalizing constants C_λ
 - 2: **for** each $i \in [n]$ **do**
 - 3: **for** each $l \in [k]$ **do**
 - 4: Compute $\omega_{l,i} = \frac{\lambda_{l,i}}{\sum_{s=1}^{k-1} \lambda_{s,i} + 1}$, if $l < k$
 - 5: Compute $\omega_{k,i} = \omega_{k,j} = 1 - \sum_{s=1}^{k-1} \omega_{s,i}$
 - 6: Compute posterior probability $p_{l,i} = \frac{\omega_{l,i} P_{X,l,i}(x_i)}{\sum_{l=1}^k \omega_{l,i} P_{X,l,i}(x_i)}$
 - 7: **end for**
 - 8: Compute labels $c_i \leftarrow \arg \max_l p_{li}$
 - 9: **end for**
 - 10: **return** Partition $\hat{I} = (c_1, \dots, c_n)$
-

Lemma 2.17. Let sample $X = (X_1, \dots, X_n) \sim P_\theta$, as given by (2.1), with $\theta \in \mathbb{L}_I$ and unknown $I = (I_1, \dots, I_k) \in \mathcal{I}_k$. Suppose k -group product-form prior, $I \sim \lambda_I^{prod}$. It follows that the component-wise posterior is expressed by,

$$p_{l*,i} = \sum_{I: i \in I_{l*}} \Pi(I|X) = \frac{\omega_{l*,i} \cdot P_{X,l*,i}(x_i)}{\sum_{l=1}^k \omega_{l,i} P_{X,l,i}(x_i)} \quad \text{for } l* \in [k], i \in [n].$$

Proof of Lemma 2.17. Using the expression of the posterior of I (given data X) as presented in Theorem 2.12 and the elementary formula (2.8), we obtain the expression

for the point-wise posterior as,

$$\begin{aligned}
p_{l*,i} &= \sum_{I:i \in I_l} \Pi(I|X) = \sum_{I:i \in I_{l*}} \frac{1}{P_X} \bigotimes_{l \in [k]} \bigotimes_{j \in I_l} \omega_{l,j} P_{X,l,j}(x_j) \\
&= \frac{\omega_{l*,i} P_{X,l*,i}(x_i)}{P_X} \sum_{I:i \in I_l} \bigotimes_{l \in [k]} \bigotimes_{j \in I_l, j \neq i} \omega_{l,j} P_{X,l,j}(x_j) \\
&\stackrel{(2.8)}{=} \frac{\omega_{l*,i} \cdot P_{X,l*,i}(x_i)}{P_X} \bigotimes_{j \in [n], j \neq i} \left(\sum_{l \in [k]} \omega_{l,j} P_{X,l,j}(x_j) \right).
\end{aligned}$$

By substituting $P_X = \prod_{i=1}^n \sum_{l=1}^k \omega_{l,i} P_{X,l,i}(x_i)$, we arrive at the final expression. $\square$

To facilitate the use of the multivariate trick and simplify the derivation of the MPAP estimator, we first present the following lemma as a preliminary step.

Lemma 2.18. *For a matrix $\alpha = (\alpha_{li}, l \in [k], i \in [n])$ and a multivariate polynomial $\Phi_\alpha(Z)$ as given in Lemma 2.14, with $Z_1 = 1$, define the multivariate polynomial,*

$$\Phi'_\alpha(Z) = \Phi_\alpha(Z) \left(\alpha_{1j} Z_1 + \alpha_{2j} Z_2 + \dots \dots + \alpha_{kj} Z_k \right)^{-1} = \prod_{\substack{i=1 \\ i \neq j}}^n \left(\sum_l \alpha_{li} Z_l \right).$$

We denote the coefficients of the monomial $Z_1^{m'_1}, \dots, Z_k^{m'_k}$ of $\Phi'_\alpha(Z)$ as,

$$A_{\mathbf{m}', -j}(a_1, \dots, a_k) = \sum_{|S|=\mathbf{m}'} \prod_{i \in S_1} \alpha_{1i} \prod_{i \in S_2} \alpha_{2i} \dots \prod_{i \in S_k} \alpha_{ki}. \quad (2.20)$$

where $S = (S_1, \dots, S_l) \in \mathcal{I}_k^{n-1}$ is a partition of the indices, $[n] \setminus i = \{1, \dots, i-1, i+1, \dots, n\}$ into k groups.

Proof. The result follows directly from Lemma 2.14. $\square$

Lemma 2.19. *Let sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ with $\theta \in \mathbb{L}_I$ and let the model as given by (2.1). Suppose cardinality-based prior $I \sim \lambda_I^{card}$. It follows that the component-wise posterior for unknown $I \in \mathcal{I}_k$ is represented as,*

$$p_{l*,i} = \frac{P_{X,l*,i}(x_i)}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} h(\mathbf{m}) A_{\mathbf{m}_{l*, -i}}(P_{X,I_1}(x), P_{X,I_2}(x), \dots, P_{X,I_k}(x)), \quad \text{for } l* \in [k], i \in [n]$$

where $\mathbf{m}_{l^*} = \mathbf{m} - e_{l^*}$ with e_{l^*} the standard basis vector (a vector with a 1 in the l^* -th position and zeros elsewhere) and $A_{\mathbf{m}_{l^*}, -i}$ the coefficient of the multivariate polynomial spanned by $n - 1$ points as given by (2.20).

Proof. Given the posterior from the bayesian analysis, Theorem 2.15, it follows that,

$$\begin{aligned} p_{l^*, i} &= \sum_{I: i \in I_{l^*}} \Pi(I|X) = \sum_{I: i \in I_{l^*}} \frac{\lambda_I^{card}}{P_X} \bigotimes_{l \in [k]} \bigotimes_{j \in I_l} P_{X, l, j}(x_j) \\ &= \frac{1}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} \sum_{\substack{I: |I| = \mathbf{m} \\ i \in I_{l^*}}} h(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{j \in I_l} P_{X, l, j}(x_j) \\ &= \frac{P_{X, l^*, i}(x_i)}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} h(\mathbf{m}) \sum_{I: |I| = \mathbf{m}} \bigotimes_{l \in [k]} \bigotimes_{\substack{j \in I_l \\ j \neq i}} P_{X, l, j}(x_j). \end{aligned}$$

For the matrix $p = (P_{X, l, j}, l \in [k], j \in [n])$, let the multivariate polynomial, as given by Lemma 2.14 with $Z_1 = 1$. Assume $j \in I_{l^*}$ for some fixed $\mathbf{m}$, it holds that $\mathbf{m}_{l^*} = \mathbf{m} - e_{l^*}$ where, e_{l^*} is the standard basis vector,

$$(e_{l^*})_i = \begin{cases} 1 & \text{if } i = l^*, \\ 0 & \text{otherwise.} \end{cases}$$

By substituting, and using Remark 2.18, the coefficients of the polynomial $\Phi'_a(Z)$ can be expressed as,

$$A_{\mathbf{m}_{l^*}, -j}(P_{X, 1}, \dots, P_{X, k}) = \sum_{I': |I'| = \mathbf{m}_{l^*}} \bigotimes_{l \in [k]} \bigotimes_{j \in I'_l} P_{X, l, j}(x_j) = \sum_{I: |I| = \mathbf{m}} \bigotimes_{l \in [k]} \bigotimes_{\substack{j \in I_l \\ j \neq i}} P_{X, l, j}(x_j),$$

where $I' = (I_1, \dots, I_{l^*} \setminus i, \dots, I_k) \in \mathcal{I}_k^{n-1}$, the final expression is derived. $\square$

The exact derivation of the $\hat{I}_{MPAP}$ under the cardinality-based prior assumption is given by, Algorithm 4.

Although the cardinality-based prior reduces the exponential complexity of the general case, computing $\hat{I}_{MPAP}$ still requires generating $n+1$ polynomials. This makes Algorithm 4 computationally demanding. Therefore, the rest of this work focuses on

4. Cardinality-based prior: Point-wise posterior

Input: Data $x_1, \dots, x_n \sim P_\theta$, prior means $\mu_1, \dots, \mu_k$, prior $\theta \sim \Pi_\theta$ and cardinality weights $g_1, \dots, g_s$ with $\sum g = 1$ ($\mathbf{m} \rightarrow g(\mathbf{m})$)

Output: Partition Estimator $\hat{I}$ // w.r.t. MPAP

```
1: for each  $l \in \{1, \dots, k\}$  do
2:   for each  $i \in \{1, \dots, n\}$  do
3:     Compute the marginal likelihood  $\alpha_{li} \leftarrow P_{X,l,i}(x_i)$ 
4:   end for
5: end for
6: Compute polynomial  $Z \rightarrow P[a_1, \dots, a_k](Z)$  // Lemma 2.14
7: Extract powers of monomials of P to get cardinalities space  $\mathcal{M}$ 
8: for each cardinality  $\mathbf{m} \in \mathcal{M}$  do
9:   Assign prior weight  $g_{\mathbf{m}}$  to  $\mathbf{m}$ 
10:  Extract coefficients  $A_{\mathbf{m}}$ 
11: end for
12: Compute  $P_X = \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}}(P_{X,1}(x), P_{X,2}(x), \dots, P_{X,k}(x))$ 
13: for each  $i \in [n]$  do
14:   Compute polynomial  $Z \rightarrow P'[a_1, \dots, a_k]_{-i}(Z)$ 
15:   Extract powers of monomials of P' to get cardinalities space  $\mathcal{M}'$ 
16:   for each  $l \in [k]$  do
17:     Compute  $A_{\mathbf{m}_l, -i}(P_{X,I_1}(x), P_{X,I_2}(x), \dots, P_{X,I_k}(x))$  // Lemma 2.19
18:     Assign  $g_{\mathbf{m}_l}$  to  $\mathbf{m}_l$ 
19:     Compute  $p_{l,i} \leftarrow \frac{P_{X,I_l,i}(x_i)}{P_X} \sum_{\mathbf{m}_l \in \mathcal{M}'} g(\mathbf{m}_l) A_{\mathbf{m}_l, -i}(P_{X,I_1}(x), P_{X,I_2}(x), \dots, P_{X,I_k}(x))$ 
20:   end for
21:   Compute labels  $c_i = \arg \max_l p_{l,i}$ 
22: end for
23: return Partition  $\hat{I} = (c_1, \dots, c_n)$ 
```

the product-form prior, which is more practical for computation.

In addition to the MPAP estimator, repeated sampling is utilized to reflect the uncertainty in the posterior distribution and to support more robust inference.

Repeated Sampling

For a given sample x , we assume the matrix $\mathbf{p} \in \mathbb{R}^{n \times k}$, where each entry $\mathbf{p} = (p_{i,j})$ represents the point-wise posterior probabilities interpreted as "the probability of the i -th data point belongs to cluster j ". To incorporate the uncertainty

inherent in these probabilistic assignments, for each x_i we perform repeated stochastic sampling from the posterior distribution. Specifically, for each x_i , we generate B independent samples $z_i^1, \dots, z_i^B \sim \text{Categorical}(\mathbf{P}_i)$. The final cluster assignment is then determined via majority voting,

$$\hat{z}_i = \arg \max_{j \in \{1, \dots, k\}} \sum_{s=1}^S \mathbb{I}(z_i^{(s)} = j),$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. This procedure yields a robust estimate of the most probable cluster label for each observation, while explicitly accounting for uncertainty in the posterior distribution.

This procedure is appropriate when direct computation of the matrix $\mathbf{p}$ is infeasible or challenging, but generating samples from it is relatively straightforward.

2.7 Example: Normal Data

To reduce computational complexity and facilitate analytical tractability, we leverage conjugate priors by assuming that both the data and the prior on θ follow Normal distributions. Specifically, we consider the following hierarchical model,

$$X_i | \theta_i, I \sim \mathcal{N}(\theta_i, 1), \quad \forall i \in [n],$$

$$\theta_i | I \sim \Pi_{I,i} = \mathcal{N}(\mu_l, 1), \quad i \in I_l,$$

$$I \sim \lambda_I,$$

where $\mu_1, \dots, \mu_k$ denote the prior means corresponding to each cluster I_l . The joint prior distribution on $\theta = (\theta_1, \dots, \theta_n)$ is then given by $\theta \sim \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I$, which, under the product-form prior assumption, simplifies to $\theta_i \sim \sum_{l=1}^k \omega_{l,i} \Pi_{l,i}$, $\forall i \in [n]$, where $\omega_{l,i}$ denotes the weight associated with assigning θ_i to group l .

Due to conjugacy, the posterior distribution for each θ_i conditional on its cluster I_l and the observed data X_{I_l} remains Normal,

$$\theta_i | X_{I_l}, I_l \sim \mathcal{N}\left(\frac{\sum_{j \in I_l} X_j + \mu_l}{m_l + 1}, \frac{1}{m_l + 1}\right),$$

where $m_l = |I_l|$ is the size of cluster l .

2.8 Quality Measures for Performance Assessment

To provide a comprehensive assessment of clustering quality, we employ three measures to compare the true partition $I = (I_1, \dots, I_k)$, $I \in \mathcal{I}_k$, with the estimated partition $\hat{I} = (\hat{I}_1, \dots, \hat{I}_s)$, $\hat{I} \in \mathcal{I}_s$, as described by Meilă [15]. Specifically, we use the *Adjusted Rand Index (ARI)* to evaluate statistical agreement, the *Variation of Information (VI)* to quantify information-theoretic dissimilarity, and the *Perfect Match Distance (PMD)* to assess exact label correspondence.

2.8.1 Adjusted Rand Index

The Rand Index (RI), introduced by Rand [20], quantifies the similarity between two clusterings I and $\hat{I}$ by measuring the agreement by counting pairs of data points.

Definition 2.20 (Rand Index). Let $I = (I_1, \dots, I_k)$ be the ground truth clustering and $\hat{I} = (\hat{I}_1, \dots, \hat{I}_s)$ be the predicted clustering. Define the contingency table $[n_{ij}]$ by

$$n_{ij} = |I_i \cap \hat{I}_j|, \quad (2.21)$$

which counts the number of data points shared by cluster I_i in the true clustering and cluster $\hat{I}_j$ in the predicted clustering, for $i = 1, \dots, k$ and $j = 1, \dots, s$. The Number of (Unordered) Object Pairs are of the following Four Types,

$$\begin{aligned} \text{(i)} \quad a &= \sum_{i=1}^k \sum_{j=1}^s \binom{n_{ij}}{2} \\ \text{(ii)} \quad b &= \sum_{i=1}^k \binom{\sum_{j=1}^s n_{ij}}{2} - \sum_{j=1}^s \binom{n_{ij}}{2} \\ \text{(iii)} \quad c &= \sum_{j=1}^s \binom{\sum_{i=1}^k n_{ij}}{2} - \sum_{i=1}^k \binom{n_{ij}}{2} \end{aligned}$$

$$(iv) \ d = \binom{n}{2} - a - b - c$$

The Rand Index is defined by,

$$\mathcal{R}(I, \hat{I}) = \frac{a + d}{\binom{n}{2}}. \quad (2.22)$$

Here, a denotes the number of point pairs that are assigned to the *same* cluster in both partitions (i), while d is the number of point pairs assigned to *different* clusters in both partitions (iv). The value b represents the number of pairs that are clustered together in the true partition but placed in different clusters in the predicted one (ii), whereas c is the number of pairs that are in different clusters in the true partition but grouped together in the predicted one (iii).

Remark 3. The Rand Index ranges from 0, indicating no agreement, to 1, indicating perfect agreement. However, as observed by [11], the Rand Index tends to concentrate in a small interval near 1 and does not fully utilize the $[0, 1]$ range.

This motivates the introduction of an adjusted version of the index.

Definition 2.21 (Adjusted Rand Index). Let Rand Index $\mathcal{R}$, as given by Definition 2.20. The Adjusted Rand Index (ARI) is defined as

$$\mathcal{AR}(I, \hat{I}) = \frac{\mathcal{R}(I, \hat{I}) - \mathbb{E}[\mathcal{R}]}{1 - \mathbb{E}[\mathcal{R}]}, \quad (2.23)$$

where $\mathbb{E}[\mathcal{R}]$ is the expected Rand Index under a random model.

Remark 4. While Rand Index does not depend on any statistical assumption of the clusterings, *ARI* uses a statistical correction via $E(\mathcal{R})$ to account for chance agreement.

Remark 5. Values of $\mathcal{AR}$ close to 1 indicate high agreement, while values near 0 suggest randomness. In case of a completely random clustering, it can be the case that $\mathcal{R} < \mathbb{E}(\mathcal{R})$ and thus the Adjusted Rand Index takes values between -1 and 0 indicating performance worse than random.

Lemma 2.22. Let $I = (I_1, \dots, I_k)$ be the ground truth clustering and $\hat{I} = (\hat{I}_1, \dots, \hat{I}_s)$ be the predicted clustering and the contingency table $[n_{ij}]$ as given by (2.21). Based on this table,

define the marginal sums

$$\begin{aligned} a_i &= \sum_{j=1}^s n_{ij}, \quad \text{the size of cluster } I_i, \\ b_j &= \sum_{i=1}^k n_{ij}, \quad \text{the size of cluster } \hat{I}_j, \end{aligned} \tag{2.24}$$

Then, the Adjusted Rand Index (ARI) between I and $\hat{I}$ can be expressed as

$$\mathcal{AR}(I, \hat{I}) = \frac{\sum_{i=1}^k \sum_{j=1}^s \binom{n_{ij}}{2} - \sum_{i=1}^k \binom{a_i}{2} \sum_{j=1}^s \binom{b_j}{2} / \binom{n}{2}}{\frac{1}{2} \left(\sum_{i=1}^k \binom{a_i}{2} + \sum_{j=1}^s \binom{b_j}{2} \right) - \sum_{i=1}^k \binom{a_i}{2} \sum_{j=1}^s \binom{b_j}{2} / \binom{n}{2}}.$$

Proof. Using the contingency table, Hubert et al. [12] reformulate the Rand Index as $\mathcal{R} = 1 + \sum_{i,j} n_{ij}^2 / \binom{n}{2} - \frac{1}{2} (\sum_i a_i^2 + \sum_j b_j^2) / \binom{n}{2}$. They further show that the expected value of this expression is $\mathbb{E}(\mathcal{R}) = 1 + 2 \sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} / \binom{n}{2}^2 - \sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} / \binom{n}{2}$. Evaluating these expressions together with (2.23) leads to the required outcome. $\square$

2.8.2 Variation of Information

The Variation of Information is an information-theoretic measure that characterizes the information content of individual clusterings as well as the shared information between them [10]. The information contained in an individual clustering is quantified by its *entropy* (in bits).

Definition 2.23. Let X be a discrete random variable representing the cluster assignments of a data point under clustering $I \in \mathcal{I}_k$. The entropy of X is denoted by $H(I)$ and is defined as follows,

$$H(I) = - \sum_i \mathbb{P}(X = i) \log \mathbb{P}(X = i) \tag{2.25}$$

Assuming $\mathbb{P}(Y = i) = p_i/n$, the probability that a randomly selected data point

belongs to cluster $i \in [k]$ of size p_i the entropy becomes

$$H(I) = - \sum_i \frac{p_i}{n} \log \left(\frac{p_i}{n} \right).$$

The information one clustering contains about the other is quantified by the *Mutual Information*, between clusterings I and $\hat{I}$

Definition 2.24. Let X and Y be discrete random variables representing the cluster assignments of a data point under two different clusterings I and $\hat{I}$ respectively. the *Mutual Information*, between clusterings I and $\hat{I}$ and is defined as follows,

$$I(I, \hat{I}) = \sum_{i,j} \mathbb{P}(X = i, Y = j) \log \frac{\mathbb{P}(X = i, Y = j)}{\mathbb{P}(X = i) \mathbb{P}(Y = j)}, \quad (2.26)$$

Assuming uniformity, $\mathbb{P}(X = i, Y = j) = n_{ij}/n$, the joint distribution of X, Y given contingency table $[n_{ij}]$ as given by (2.21) in Lemma 2.22 with cluster sizes a_i, b_j for I and $\hat{I}$ given by (2.24) respectively, the expression transforms into,

$$I(I, \hat{I}) = - \sum_{i=1}^k \sum_{j=1}^s \frac{n \cdot n_{ij}}{a_i b_j} \log \left(\frac{n \cdot n_{ij}}{a_i b_j} \right)$$

Definition 2.25 (Variation of Information [15]). The *Variation of Information* (VI) between two partitions $I \in \mathcal{I}_k$ and $\hat{I} \in \mathcal{I}_s$ is defined as,

$$VI(I, \hat{I}) = H(I) + H(\hat{I}) - 2I(I, \hat{I})$$

where $H(\cdot)$ denotes the entropy of a clustering, and $I(\cdot, \cdot)$ is the mutual information between two clusterings as, given by (2.25) and (2.26) respectively. The expanded form of the Variation of Information is given by,

$$VI(\mathbf{c}, \hat{\mathbf{c}}) = - \sum_i \frac{a_i}{n} \log \left(\frac{a_i}{n} \right) - \sum_j \frac{b_j}{n} \log \left(\frac{b_j}{n} \right) - 2 \sum_{i,j} \frac{n_{ij}}{n} \log \left(\frac{n_{ij} \cdot n}{a_i b_j} \right).$$

Remark 6. As shown in [15], VI is an n -invariant metric meaning that it is better understandable to human and it only depends on the relative sizes of the clusters.

Remark 7. VI is bounded by $VI(I, \hat{I}) \leq 2 \cdot \log n$ and given K^* an upper bound of the

number of clusters of the two clusterings then it also holds that $VI(I, \hat{I}) \leq 2 \cdot \log K^*$, with $K^* \leq \sqrt{n}$.

For easier interpretation, we use a normalized version of the Variation of Information (VI), defined as

$$AVI = \frac{VI}{2 \log K^*},$$

where $K^* = \max\{k, s\}$ is the larger number of clusters in either I or $\hat{I}$. A value of AVI close to 1 indicates high similarity between clusterings, while values near 0 indicate dissimilarity.

2.8.3 Set Matching criterion

The *set matching criterion*, introduced by Meilă and Heckerman [16], is defined as

$$\mathcal{H}(I, \hat{I}) = 1 - \frac{1}{n} \max_{\pi \in S_k} \sum_{i=1}^k n_{i, \pi(i)},$$

where $n_{ij} = |I_i \cap \hat{I}_j|$ denotes the confusion matrix between the true clustering $I = (I_1, \dots, I_k)$ and the predicted clustering $\hat{I} = (\hat{I}_1, \dots, \hat{I}_s)$, and $\pi \in S_k$ is a permutation over k cluster indices. This criterion quantifies the *minimum fraction of mismatched data points* under the best possible one-to-one alignment between clusters.

Finding the optimal permutation π corresponds to solving the well-known *assignment problem* in combinatorial optimization. Rather than exhaustively searching all $k!$ permutations, the problem can be solved efficiently using the *Hungarian algorithm*, which finds the maximum-weight matching in a bipartite graph [16, 18, 19].

Perfect Match Score

In this thesis, an adaptation of the set matching criterion is used, referred to as the *Perfect Match Score (PMS)*. This reformulation is chosen for interpretation purposes; in order to align with the Adjusted Rand Index (ARI). PMS measures the proportion of data points correctly matched under the optimal one-to-one alignment of clusters.

Definition 2.26 (Perfect Match Score). Let $I = (I_1, \dots, I_k) \in \mathcal{I}_k$ and $\hat{I} = (\hat{I}_1, \dots, \hat{I}_s) \in \mathcal{I}_s$ be two clusterings and let the confusion matrix $n_{ij} = |I_i \cap \hat{I}_j|$ denote the number of shared

data points between clusters I_i and $\hat{I}_j$. The set matching criterion is defined as,

$$PMS = \frac{1}{n} \max_{\pi \in S_k} \sum_{i=1}^k n_{i, \pi(i)}$$

where S_k is the set of all permutations of $\{1, \dots, k\}$.

Remark 8. The Set Matching Criterion and Perfect Match Score are permutation-based measures that rely solely on the combinatorial alignment of cluster labels, without making any statistical assumptions about the underlying clustering process. This contrasts with measures such as Variation of Information (VI), which treat cluster labels as random variables, and the Adjusted Rand Index (ARI), which accounts for chance agreement through an adjustment based on the expected Rand index, $E(\mathcal{R})$.

Example 2.27. Suppose that the ground truth, $I = [0, 0, 1, 1, 2, 2]$ and the predicted $\hat{I} = [1, 1, 2, 2, 0, 0]$. Each clustering has 3 clusters of equal size. The resulting contingency table is given by,

	$\hat{I}_0$	$\hat{I}_1$	$\hat{I}_2$
I_0	0	2	0
I_1	0	0	2
I_2	2	0	0

The quality measures in this case result in, $PMS = \frac{6}{6} = 1$, $AVI = 1$, $ARI = 1$ indicating perfect agreement between the two clusterings.

Example 2.28. Suppose that the ground truth, $I = [0, 0, 1, 1, 2, 2]$ and the predicted $\hat{I} = [0, 1, 0, 1, 2, 2]$. The resulting contingency table is the following,

	$\hat{I}_0$	$\hat{I}_1$	$\hat{I}_2$
I_0	1	1	0
I_1	1	1	0
I_2	0	0	2

The quality measures in this case are evaluated as, $PMS = \frac{4}{6}$, $AVI = 0.4177$ and $ARI \approx 0.167$. Meaning that only 4 out of 6 points are correctly matched based on PMS, the information content differs significantly between the clusterings (ARI) and weak agreement after accounting for random label alignment (AVI).

3 Simulation study

So far we considered n 1-dimensional observations $X_1, \dots, X_n \in \mathbb{R}$ with 1-dimensional parameters $\theta_1, \dots, \theta_n \in \mathbb{R}$ and developed our method for this case. However, it is possible to extend to a higher-dimensional case. In the below simulation study, we consider the case of 2-dimensional observations $X_i \in \mathbb{R}^2$ and corresponding parameters $\theta_i \in \mathbb{R}^2$. Consequently, the likelihood P_{θ_i} for X_i and the prior $\Pi_{I,i}$ for θ_i are all bivariate.

3.1 Dataset Design

To generate the simulation dataset, we begin by specifying the *true parameters*: the total number of data points n , the number of clusters k , the true partition I , and the true cluster means $\mu = (\mu_1, \mu_2)$. The values of n and k are selected according to Table 3.1, which is designed to capture a diverse range of clustering scenarios—from low sample regimes ($n = 50$) to large-scale settings ($n = 2000$). The number of clusters varies from $k = 3$ to $k = 11$, introducing different levels of noise and complexity into the data.

Once these values are fixed, the remaining components—namely, the true cluster means and the true partition—are determined by specific procedures, described in the following subsections.

Parameter	Values
Dimensions (d)	2
Clusters (k)	3, 5, 7, 11
δ	0.2, 0.84, 1.49, 2.13, 2.78, 3.42, 4.07, 4.71, 5.36, 6.0
Sample Size (n)	50, 100, 250, 500, 750, 1000, 1250, 1500, 1750, 2000

Table 3.1: Parameter grid for the simulation study.

3.1.1 True Partition and Cluster Means

The true partition I is constructed by sampling with replacement from the set $[n]$ (see Appendix A.1, Listing A.1). The key challenge in this setting is to generate cluster means that are *well-separated*—ensuring a minimum pairwise distance of δ between any two cluster centers—while allowing for a broad variety of spatial configurations.

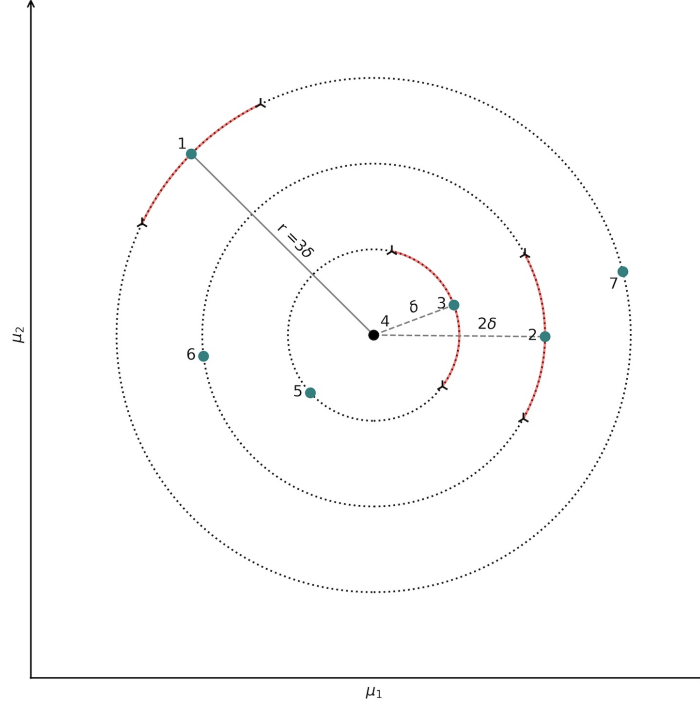


Figure 3.1: Design of cluster means according to Equation (3.1).

Generation of Cluster Means

To address this challenge, we fix the mean of the central cluster without loss of generality as $\mu_{\lfloor k/2 \rfloor} = [1, 1]$. The remaining cluster centers are then positioned on concentric circles centered at $[1, 1]$, with radii scaled proportionally to δ . Formally, each cluster mean satisfies,

$$(\mu_{l,1} - 1)^2 + (\mu_{l,2} - 1)^2 = (r \cdot \delta)^2, \quad \text{where} \quad r = \left\lceil l - \left\lfloor \frac{k}{2} \right\rfloor \right\rceil. \quad (3.1)$$

In this setup, at most two points are sampled per circle to ensure sufficient angular separation. To guarantee that all pairwise distances between cluster means are at least δ , the second point on each circle is drawn only if it lies outside the exclusion

zone of radius δ from the first. This rejection sampling procedure defines the feasible region for selecting the second point, as illustrated in Figure A.1.

Example 3.1. Assume $k = 7$, with the fourth cluster mean fixed at $[1, 1]$. The cluster means for groups 1, 2, and 3 are placed on circles of radius 3δ , 2δ , and δ , respectively. Similarly, clusters 5, 6, and 7 are placed on concentric circles of radii δ , 2δ , and 3δ , respectively. To enforce a minimum pairwise separation of δ between the cluster means, the following constraints are applied,

$$|\mu_3 - \mu_5| > \delta, \quad |\mu_2 - \mu_6| > \delta, \quad |\mu_1 - \mu_7| > \delta.$$

Dataset Generation

Given the total sample size n and the cluster sizes n_l derived from the true partition, we sample each data point $X_{i,l}$ independently from a bivariate normal distribution centered at the cluster mean μ_l as follows:

$$X_{i,l} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(\begin{bmatrix} \mu_{l,1} \\ \mu_{l,2} \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right), \quad (3.2)$$

where $i \in I_l$ and $l \in [k]$.

Remark 9. Since the covariance matrix is fixed to the identity, the parameter δ directly controls the degree of separation between clusters (see Figure A.1).

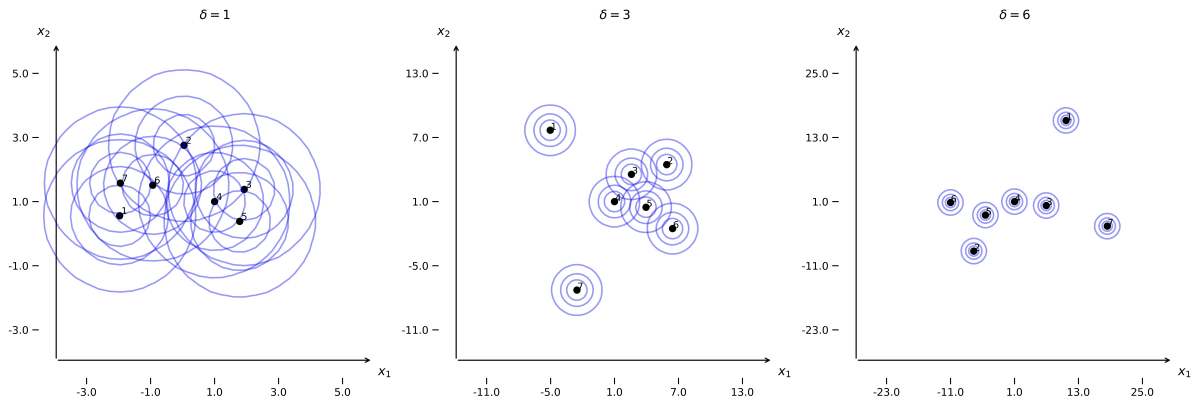


Figure 3.2: Samples drawn from bivariate normal distributions as defined by Equation (3.2), for varying values of δ .

The parameter δ ranges from very small values (e.g., 0.2), where clusters exhibit significant overlap, to large values (e.g., 6.0), where clusters are well-separated. This range enables us to evaluate the algorithm’s sensitivity to noise and its ability to recover underlying cluster structures under different levels of separation.

3.2 Simulation Study - Results

3.2.1 Effect of the Number of Clusters

Figure 3.3 shows how the number of estimated clusters $\hat{k} \in \{3, 5, 7, 11\}$ impacts the performance of the ProdForm algorithm ¹ under an upper bound $k = 11$. Performance is measured using the Adjusted Rand Index (ARI) across different values of the separation parameter δ and sample sizes $n = 50, 500$, and 1750 .

Performance generally improves as δ increases, indicating that the algorithm more accurately recovers the true cluster structure when clusters are better separated. However, at smaller sample sizes, the performance trend becomes less stable, with more pronounced fluctuations across different values of $\hat{k}$. This variability is likely due to the limited data, which increases the algorithm’s sensitivity to noise and outliers.

As the sample size increases (e.g., $n = 500$ and 1750), the relationship between δ and performance becomes more consistent and exhibits a clearer monotonic trend. Nonetheless, certain irregularities persist. For instance, notable drops in performance occur at $n = 500, \hat{k} = 7, \delta = 3.42$ and $n = 50, \hat{k} = 5, \delta = 5.36$. These deviations are likely driven by outliers or unfavorable data configurations that can hinder accurate clustering, particularly in low-sample settings.

Notably, for larger sample sizes—particularly at $n = 750$ —the performance for $\hat{k} = 7$ and $\hat{k} = 11$ tends to surpass that of $\hat{k} = 3$ and $\hat{k} = 5$. This observation may be attributed to the cluster-to-size ratio: with fewer clusters, each cluster absorbs more points, potentially increasing within-cluster variability and the influence of outliers. Additionally, given that the algorithm operates under an upper bound of $k = 11$, it may be less effective at accurately estimating a small number of clusters, possibly

¹Prediction: In the following analysis, the cluster centers obtained from KMeans are used to initialize the algorithm, serving as an empirical approach.

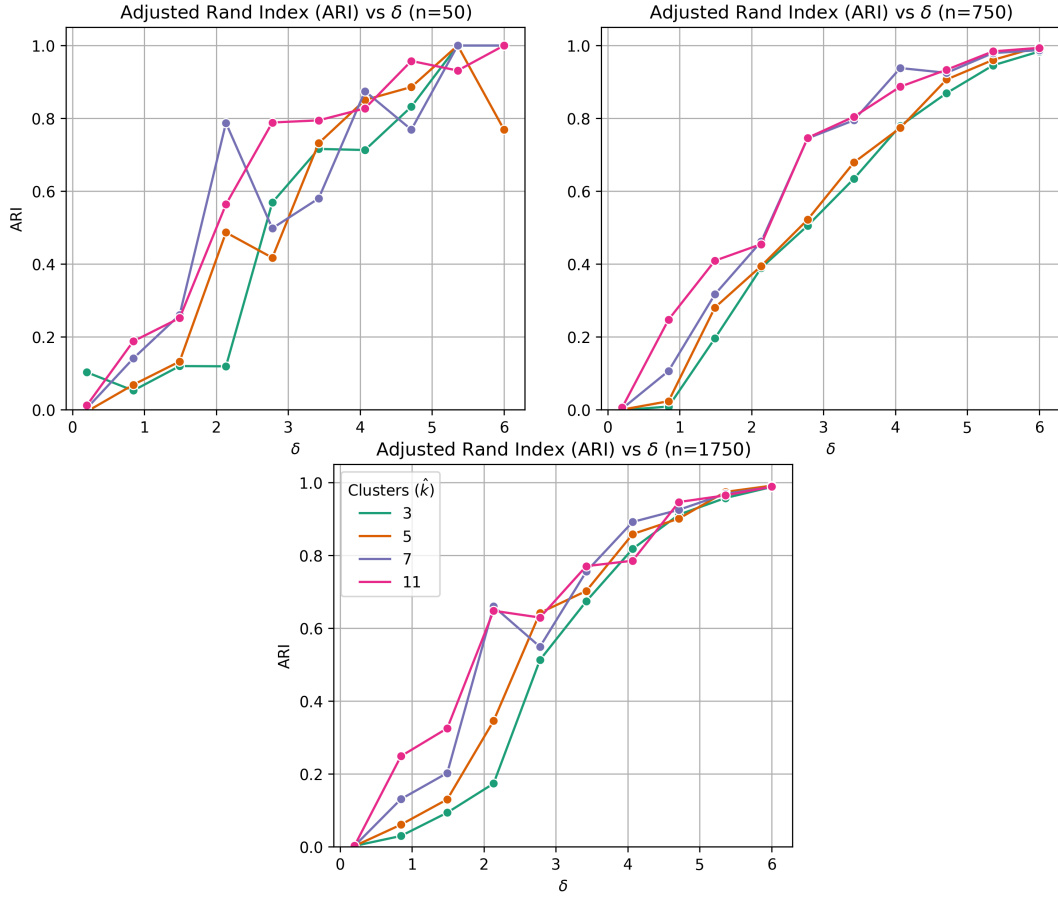


Figure 3.3: ProdForm performance (ARI) as a function of δ , across cluster and sample sizes.

misallocating points when the true number is substantially lower.

3.2.2 Effect of the Separation Parameter δ

Figures 3.3 and 3.4 illustrate that the performance of the algorithm deteriorates at lower values of the separation parameter δ , as measured by both ARI and Adjusted Variation of Information (Adj. VI). This degradation in performance is attributable to the increased overlap between clusters, which complicates the accurate recovery of the true partition structure.

An increasing trend in performance is observed as δ increases, with the algorithm achieving near-optimal scores (≈ 1) at $\delta = 6$ for all considered sample sizes. At small δ values, particularly when the sample size is limited, the insufficient number of data points impairs the algorithm's ability to adequately characterize cluster shapes. Con-

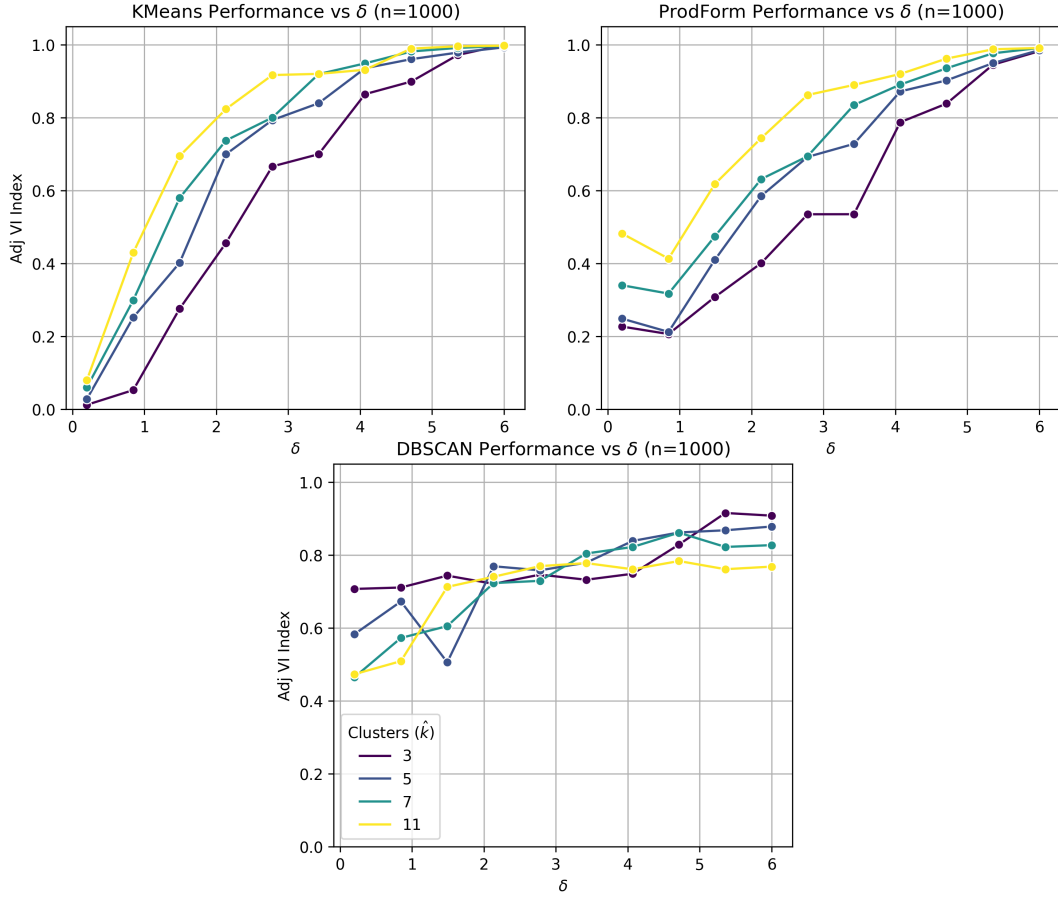


Figure 3.4: Performance measured by Adjusted Variation of Information (Adj. VI) for KMeans, ProdForm, and DBSCAN at sample size $n = 1000$.

sequently, ProdForm exhibits greater variability in its estimates and reduced accuracy in identifying the underlying partition.

With larger sample sizes, the performance trend becomes more pronounced and exhibits reduced variability, reflecting the algorithm's improved capacity to recover the true clustering structure when clusters are well separated.

3.2.3 Performance Comparison with KMeans and DBSCAN

Figure 3.4 and Table 3.2 presents a comparative analysis of the ProdForm algorithm against KMeans and DBSCAN for a sample size of $n = 1000$ and $n = 2000$, respectively. Overall, ProdForm exhibits performance that is broadly comparable to both benchmark methods on average (w.r.t. cluster size). However, in scenarios with overlapping clusters, ProdForm consistently outperforms KMeans. This advantage

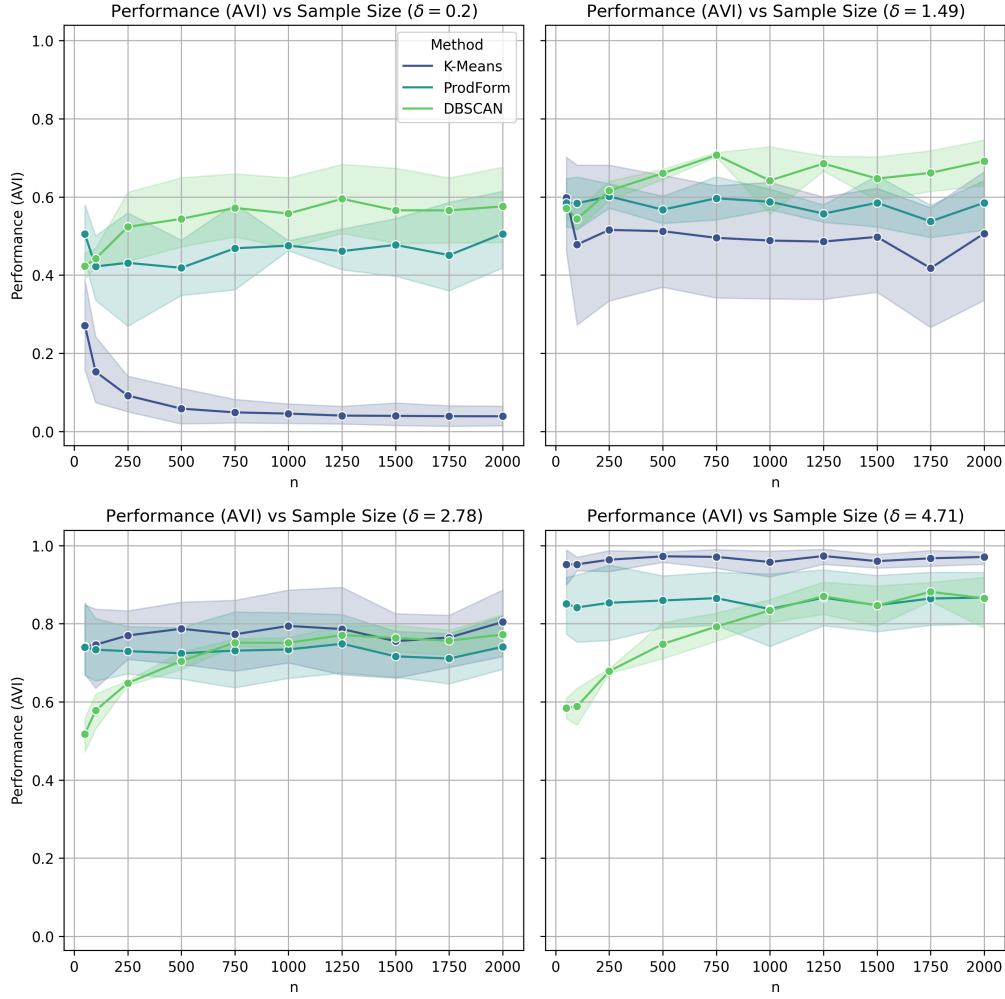


Figure 3.5: Performance measured by AVI of KMeans, ProdForm, and DBSCAN against the sample size.

is likely due to ProdForm’s probabilistic modeling framework, which accounts for uncertainty and enables more accurate identification of ambiguous or overlapping points—an area where KMeans typically underperforms.

DBSCAN exhibits relatively stable performance across different cluster configurations, with less variability compared to KMeans and ProdForm. Its performance remains within a narrower range, suggesting robustness to changes in sample size and separation. At lower values of δ , DBSCAN outperforms both KMeans and ProdForm, though it only modestly exceeds random assignment levels. However, as the separation between clusters increases, its performance plateaus and becomes notably lower than that of KMeans and ProdForm, both of which show more substantial im-

provements under well-separated conditions.

As shown in Figure 3.5, the performance of all methods improves as the separation parameter δ increases. At low values of δ , ProdForm outperforms KMeans but trails behind DBSCAN. Once the clusters become well-separated, the performance differences among the algorithms diminish. Under these conditions, ProdForm reaches an ARI of approximately 0.7, which increases to around 0.9 for larger δ , demonstrating its capacity to achieve high accuracy when the clustering structure is clear.

δ	KMeans			ProdForm			DBSCAN		
	ARI	AVI	PMS	ARI	AVI	PMS	ARI	AVI	PMS
0.844	0.260	0.455	0.425	0.107	0.458	0.298	0.001	0.570	0.117
2.133	0.797	0.841	0.893	0.559	0.766	0.738	0.192	0.774	0.396
6.000	0.998	0.997	0.999	0.989	0.990	0.995	0.695	0.872	0.768

Table 3.2: Performance comparison of KMeans, ProdForm, and DBSCAN for varying values of δ at $n = 2000$, $\hat{k} = 11$. Metrics reported are ARI, AVI, and PMS.

4 Real data and application

4.1 The Iris Flower Dataset

The Iris dataset is a classic and widely used benchmark in pattern recognition and machine learning. It consists of 150 samples from three Iris species: *Setosa*, *Verisicolor*, and *Virginica*. Each sample is described by four continuous numerical features: `sepal_length`, `sepal_width`, `petal_length`, and `petal_width`, measured in centimeters (*cm*). The goal is to uncover the inherent cluster structure corresponding to the flower species based solely on these measurements. The target labels are not used during clustering and are reserved solely for evaluating performance.

Feature Aggregation

To reduce dimensionality and enable clustering in a two-dimensional space, we construct two composite features by aggregating the original measurements as follows,

$$\text{sepal_area} = \text{sepal_length} \times \text{sepal_width},$$

$$\text{petal_area} = \text{sepal_length} \times \text{petal_width}.$$

These simple multiplicative transformations act as rough proxies for the physical areas of the sepal and petal, respectively measured in cm^2 . The resulting feature space is thus reduced from four dimensions to two, providing a compact and interpretable representation. Each flower sample is now represented by the two-dimensional vector as,

$$\mathbf{flower}_i = \begin{bmatrix} \text{petal_area}_i \\ \text{sepal_area}_i \end{bmatrix},$$

where $i = 1, \dots, 150$ indexes the individual flower samples. This reduced represen-

tation facilitates intuitive visualization and allows for straightforward application of clustering algorithms. Specifically, it enables the use of the Product-form Bayesian clustering method (ProdForm), leveraging the Point-wise Posterior approach described in Algorithm 3. This two-dimensional embedding also permits fair and direct comparison with classical clustering algorithms such as KMeans.

4.2 The Iris dataset - Results

4.2.1 Prior Mean Selection

A key insight in the application of the product-form Bayesian clustering algorithm is its strong sensitivity to the initial choice of prior means, denoted by μ_θ . The spatial configuration of these means has a direct influence on the geometry of the resulting clusters.

For instance, as illustrated in Figure 4.1, when prior means are initialized along the horizontal axis in the feature space, the resulting clusters tend to be horizontally aligned. Conversely, initializing them along a vertical axis leads to vertically aligned clusters. Moreover, random initialization of prior means from the dataset can produce highly variable results. In some instances, this randomness yields overlapping or poorly separated clusters—particularly when the sampled means are unrepresentative of the underlying data structure. Likewise, initializing all prior means at the origin imposes uniformity on the cluster selection process, resulting in unstructured and non-informative partitions.

This geometric bias highlights the critical role of informed initialization in the introduced Bayesian clustering model with a product-form prior, especially in low-dimensional feature representations. To provide meaningful insights, we adopt an Empirical Bayes approach by using the cluster centers obtained from the KMeans algorithm as prior means in our Bayesian model. This enables a data-driven initialization and allows for a direct comparison between the clustering results of KMeans and our proposed method.

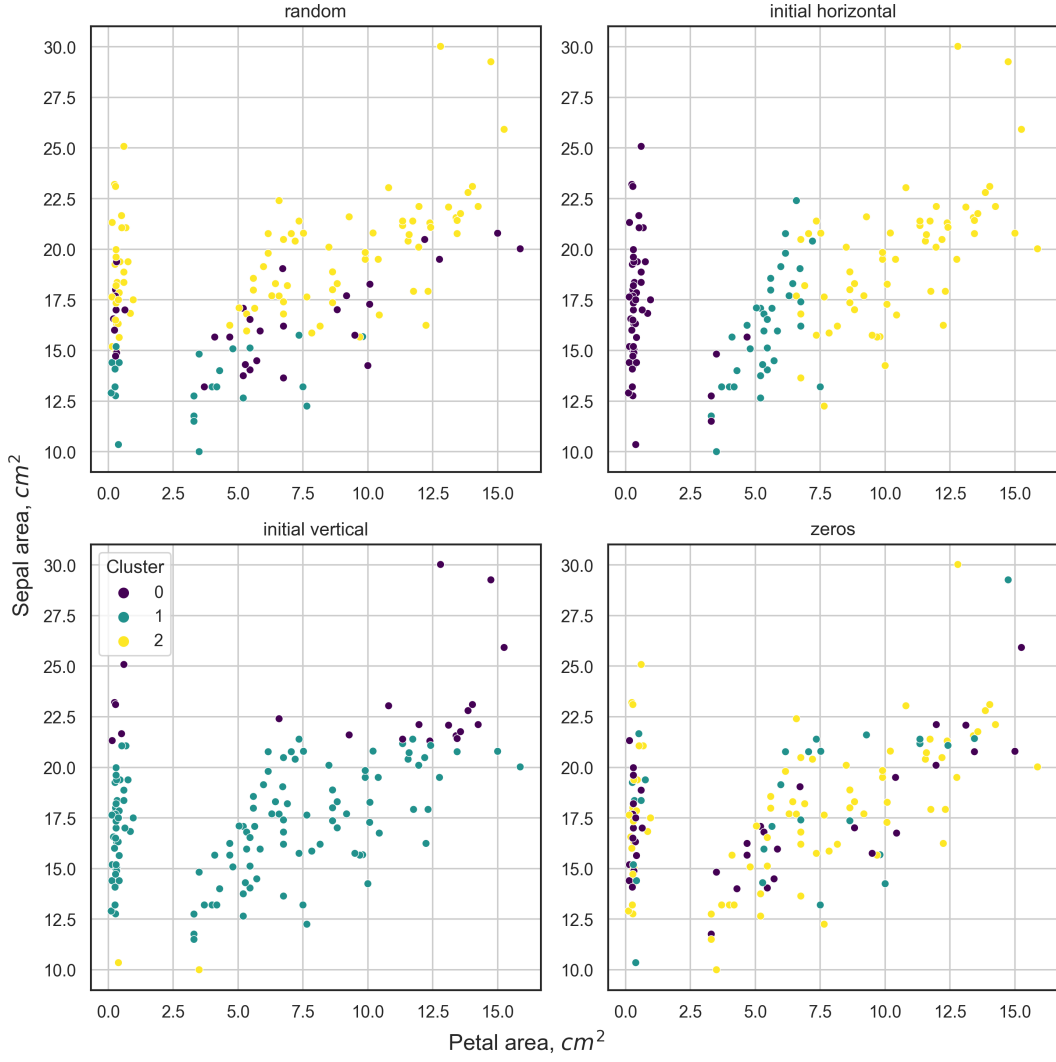


Figure 4.1: ProdForm under different prior mean initializations, $k = 3$.

4.2.2 Number of clusters

As illustrated in Figure 4.2, initializing the model with $k = 7$ clusters serves as an upper bound on the number of clusters that can emerge. While traditional methods such as KMeans impose this cluster count as a fixed constraint, the introduced Bayesian clustering model allows for a more flexible structure. Specifically, it infers the optimal number of clusters dynamically based on the posterior distribution on $\mathcal{I}_k$ since $\mathcal{I}_l \subseteq \mathcal{I}_k$ for each $l \in [k]$.

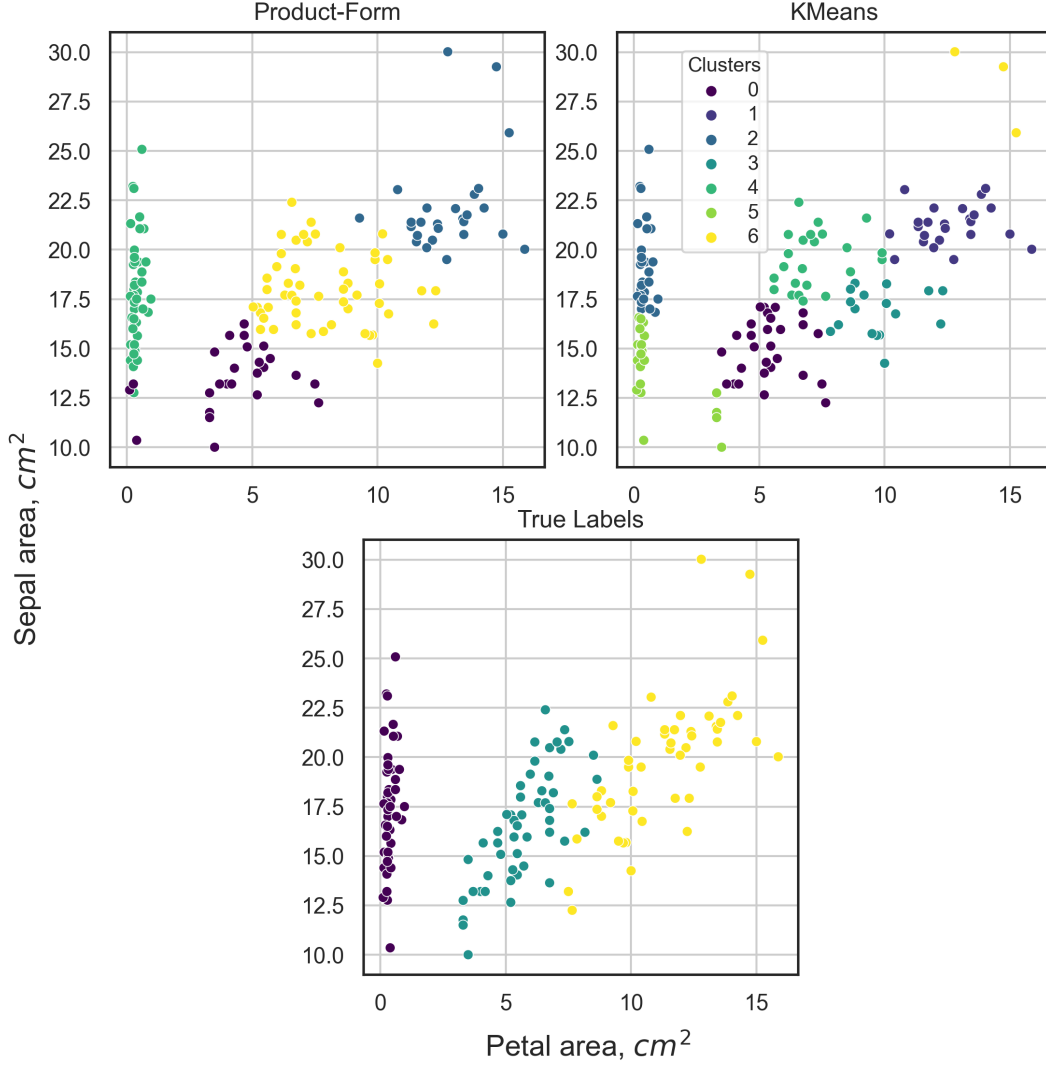


Figure 4.2: Clustering of Iris dataset samples based on petal and sepal area.

4.2.3 Model assumptions

We assume that each observation follows a normal model, $X_i \sim \mathcal{N}(\theta_i, \mathbf{I}_2)$, where $X_i \in \mathbb{R}^2$. However, due to the unsupervised nature of the clustering process, this assumption cannot be verified prior to analysis. Consequently, standard normalization techniques are not applicable, and model misspecification is, to some extent, unavoidable.

Once the partition estimate $\hat{I}_{\text{MPAP}}$ has been obtained, the validity of the underlying distributional assumptions can be assessed retrospectively, as discussed in Sec-

tion 2.7. Specifically, we assume

$$\forall i \in [n], \quad \theta_i \mid \hat{I} \sim \mathcal{N}(\mu_i, \mathbf{I}_2), \quad \text{where } \mu_i = \mu_l \text{ for } i \in \hat{I}_l.$$

The estimators are defined as follows, $\hat{I} = \hat{I}_{\text{MPAP}}$, $\hat{\theta} = \mathbb{E}_\theta[\theta \mid \hat{I}]$, and consequently,

$$\forall i \in [n], \quad X_i \mid \hat{I}, \hat{\theta} \sim \mathcal{N}(\hat{\theta}_i, \mathbf{I}_2), \quad \text{where } \theta_i = \theta_l \text{ for } i \in \hat{I}_l.$$

Covariance

The first step is to check the covariance between `sepal_area` and `petal_area`. The Pearson correlation test fails to reject the null hypothesis, indicating that there is no significant correlation between the two variables. Therefore, the bivariate normal distribution can be considered as the product of two univariate distributions.

Cluster	Pearson r	p-value	Significant
0	0.2607	0.2296	No
1	0.3097	0.1409	No
2	0.2836	0.0534	No
3	0.1461	0.2827	No

Table 4.1: Pearson Correlation Results by Cluster

Normality

Conveniently, the bivariate distribution can be decomposed into the product of two univariate distributions for `petal_area` and `sepal_area`, respectively. To assess the normality of each feature across the four clusters, standard techniques were employed, including the Shapiro–Wilk test, the Kolmogorov–Smirnov (KS) test, and Q–Q plots. To control for the increased risk of Type I errors (false positives) due to multiple hypothesis testing, the significance level was adjusted using the Bonferroni correction. As eight tests were conducted in total, the corrected significance threshold is given by $\alpha_{\text{new}} = \frac{0.05}{8}$.

Cluster	Feature	KS-statistic	p-value	Normal ($p > 0.00625$)
0	Sepal Area	0.0945	0.9741	Yes
1	Sepal Area	0.2492	0.0846	Yes
2	Sepal Area	0.0866	0.8426	Yes
3	Sepal Area	0.1019	0.5710	Yes
0	Petal Area	0.1911	0.3275	Yes
1	Petal Area	0.0727	0.9985	Yes
2	Petal Area	0.2660	0.0020	No
3	Petal Area	0.1214	0.3524	Yes

Table 4.2: Kolmogorov–Smirnov test results for normality of sepal and petal area distributions within each cluster.

Overall, the assumption of normality appears reasonable across clusters. However, special attention should be given to the petal area distribution of θ_{21} of the second cluster, where a deviation from normality is indicated. A summary of the KS test results is provided in Table 4.2, while additional diagnostic plots and test outcomes are available in Appendix A.3.

4.2.4 Distribution of Clusters

We consider each cluster individually under a two-step Hierarchical Bayesian framework. For a fixed cluster $l \in [k^*] \subseteq \{1, \dots, k\}$, the following model is assumed,

$$\theta \mid \hat{I} \sim \mathcal{N}(\mu_l, \mathbf{1}),$$

$$X \mid \hat{I}, \hat{\theta} \sim \mathcal{N}(\hat{\theta}_l, \mathbf{1}),$$

where $\hat{I}$ denotes the estimated partition under the MPAP. Given the conjugate nature of the model, the posterior mean μ_l is analytically tractable, as described in Section 2.7. To validate the assumed variance, we perform a chi-squared test for each cluster. The null hypothesis that the cluster variance is equal to one is rejected in all cases. As a result, the sample variance is used as an empirical estimate of the true variance within each cluster. The resulting fitted distributions per cluster are illustrated in Figure A.1.

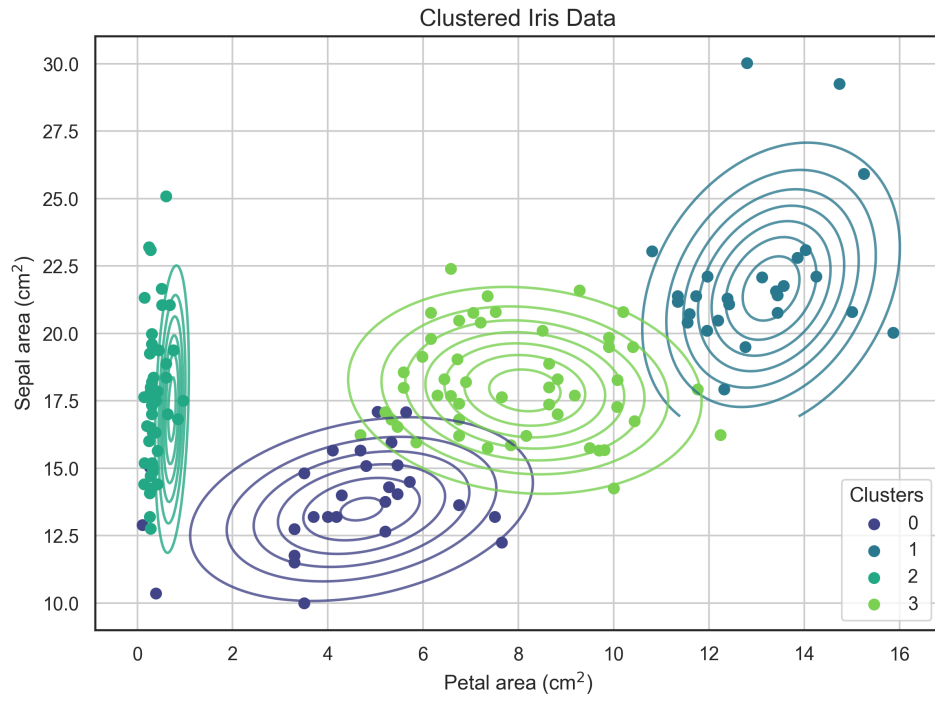


Figure 4.3: Retrieval of Distribution of Iris Data for each cluster provided the $\hat{I}_{MPAP}$ -estimate using a Product-Form prior.

5 Conclusion

5.1 Conclusion

The Bayesian clustering framework developed in this thesis builds upon the foundational model proposed by Belitser et al., which introduced a principled probabilistic approach to clustering, inspired by the classical signal-plus-noise paradigm. At the heart of this methodology lies the observation model,

$$X \sim \bigotimes_{i=1}^n P_{\theta_i},$$

where each observation X_i is generated independently from a distribution parameterized by θ_i , and clustering is imposed through a partition-based structure on the parameter vector $\theta = (\theta_1, \dots, \theta_n)$.

The key Bayesian innovation is the treatment of partitions as latent variables, over which a prior distribution is placed. This induces a hierarchical model that allows for *posterior inference* on both parameters and cluster structure. Notably, one of its strengths lies in the ability to specify an upper bound on the number of clusters, which improves model flexibility.

This work extended the original two-group model to support *arbitrary k-group clustering*, introducing two distinct types of priors over the space of partitions: the *product-form prior* and the *cardinality-based prior*. These priors offer complementary advantages. The product-form prior enables tractable inference by *factorizing* the prior probability across cluster components, i.e.,

$$\lambda_I^{\text{prod}} = C_\lambda \prod_{l=1}^{k-1} \prod_{i \in I_l} \lambda_{l,i},$$

with normalization handled analytically. On the other hand, the cardinality-based

prior defines probabilities over partitions based on their group sizes and leverages the *polynomial trick* (and its multivariate generalization) to express marginal and posterior quantities in closed form.

The Adjusted Rand Index, Variation of Information, and Perfect Match Score collectively provide complementary perspectives on clustering quality by measuring statistical agreement, information-theoretic similarity, and exact label alignment, respectively. Together, these metrics offer a robust framework for comprehensive performance assessment of clustering algorithms under varied scenarios.

While both priors yield elegant analytic expressions and enable exact posterior computations for moderate n , their *scalability* diverges in practical settings. The cardinality-based prior becomes computationally expensive as the dataset size increases, due to the combinatorial explosion in the number of partitions with fixed cardinalities. In contrast, the product-form prior retains much of the model’s probabilistic rigor while maintaining *linear computational complexity* with respect to the number of data points, making it more suitable for high-dimensional and large-scale applications. By utilizing the Point-wise a posteriori estimator $\hat{I}_{MAP}$, the product-form approach performs competitively with established methods such as KMeans, while offering the additional advantage of uncertainty quantification through posterior inference. It is further noticable that Repeat Sampling does not significantly improve performance.

The Iris analysis and simulation study highlight the Bayesian model’s flexibility in accurately determining the optimal number of clusters in practice. The performance of clustering methods is strongly influenced by the choice of feature representation and initialization strategy. Notably, the product-form Bayesian clustering approach, when initialized with KMeans cluster centers, effectively uncovers meaningful groupings that correspond well to the underlying species. However, the method also exhibits limitations. It is sensitive to the initialization of prior means and relies on distributional assumptions—specifically, normality of cluster components—which may not always hold in practice. Additionally, the selection of k , although bounded, still requires further refinement for data-driven adaptivity.

The simulation study demonstrates that the ProdForm Bayesian clustering algorithm effectively recovers cluster structures across a wide range of sample sizes, cluster counts, and separation levels. Performance improves consistently as the separation parameter δ increases, highlighting the algorithm’s sensitivity to cluster overlap.

Compared to benchmark methods such as KMeans and DBSCAN, ProdForm shows superior performance in scenarios with overlapping clusters due to its probabilistic modeling framework. However, its accuracy depends on sample size and initialization. Overall, the designed dataset and simulation results validate ProdForm’s robustness and flexibility in capturing complex clustering patterns under diverse conditions.

5.2 Future Work

Future research directions could include the development of adaptive strategies for selecting prior means, such as empirical Bayes methods or data-driven initialization heuristics. Another promising direction is improving the scalability and efficiency of posterior inference—especially in high-dimensional or non-Gaussian settings—through approximation techniques or variational methods.

Extension to Higher Dimensions

In this work, we focused primarily on univariate and bivariate observations, where $X_1, \dots, X_n \in \mathbb{R}$ or $\mathbb{R}^2$. Extending the ProdForm framework to higher-dimensional settings remains a natural and important next step.

In the multidimensional setting, $X = (X_1, \dots, X_n) \in \mathbb{R}^{n \times d}$, where d denotes the dimensionality of the feature space. The Bayesian clustering construction provided in Lemma 2.3 remains formally invariant with respect to d ; however, the dimensionality directly impacts the computation of the posterior density $P_{X,I,i}(x)$ term P_X , which becomes increasingly complex in higher dimensions, $P_{X,I,i}(x) = \int \cdots \int P_{\theta_i}(x) d\Pi_{I,i}(\theta_i)$.

While normality assumptions have allowed for analytical tractability in this work, relaxing these assumptions—by considering non-Gaussian or mixture-based models—could significantly broaden the applicability of the ProdForm approach in complex clustering scenarios. Developing model diagnostics and assumption checks tailored to clustering remains an open problem, as these assumptions cannot be verified a priori. Exploring alternative modeling frameworks for assessing data suitability in probabilistic clustering is therefore an important direction for future research.

6 Bibliography

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Appendix

A.1 Proofs: Cardinality-based prior as a probability distribution

Binary Clustering

Proposition A.1. Assume a prior λ on the partition space $\mathcal{I}_2$, $I \sim \lambda_I$ of the form $\lambda_I = c_m g(m)$ where $m \in [n]$ is the cardinality of I_1 , $I = (I_1, I_1^c)$ and $g(\cdot)$ probability distribution such that, $\sum_{m=0}^n g(m) = 1$. It must hold that $c_m = \binom{n}{m}^{-1}$.

Proof of Proposition A.1. Since $(\lambda_I, I \in \mathcal{I}_2)$ is a distribution, then $\sum_{I \in \mathcal{I}_2} \lambda_I = 1$.

$$\sum_{I \in \mathcal{I}_2} \lambda_I = \sum_{m=0}^n \sum_{I: |I|=m} c_m g(m) = \sum_{m=0}^n \binom{n}{m} c_m g(m) \quad (\text{A.1})$$

Thus, the required m - dependent constant is, $c_m = \binom{n}{m}^{-1}$, $\forall m \in [n]_0$. $\square$

k -Groups Clustering

Proposition A.2. Assume a prior λ on the partition space $\mathcal{I}_k$, $I \sim \lambda_I$ of the form $\lambda_I = c_{\mathbf{m}} g(\mathbf{m})$ where $g: \mathcal{M} \rightarrow [0, 1]$, is a nonnegative function s.t. $\sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) = 1$ with,

$$\mathcal{M} = \{\mathbf{m} = (m_1, \dots, m_k) \in \mathbb{N}_0^k : m_l = |I_l|, \forall l \in [k], I = (I_1, \dots, I_k) \in \mathcal{I}_k\}.$$

Then the constants $c_{\mathbf{m}}$ must satisfy,

$$c_{\mathbf{m}} = \left(\binom{n}{m_1, \dots, m_k} \right)^{-1}, \quad \forall \mathbf{m} \in \mathcal{M}.$$

Proof of Proposition A.2. Suppose λ_I is a prior on $\mathcal{I}_k$ of the form $\lambda_I = c_{\mathbf{m}} g(\mathbf{m})$. Abusing

notation $\mathbf{m} = |I|$ to be the vector $\mathbf{m} = (m_1 = |I_1|, \dots, m_k = |I_k|)$. Given that the number of partitions with $|I| = \mathbf{m}$ is equal to the multinomial coefficient $\binom{n}{m_1, \dots, m_k}$, it follows that,

$$\sum_{I \in \mathcal{I}_k} \lambda_I = \sum_{\mathbf{m} \in \mathcal{M}} \sum_{I: |I| = \mathbf{m}} c_{\mathbf{m}} g(\mathbf{m}) = \sum_{\mathbf{m} \in \mathcal{M}} \binom{n}{m_1, m_2, \dots, m_k} c_{\mathbf{m}} g(\mathbf{m}). \quad (\text{A.2})$$

By requiring $\sum_{I \in \mathcal{I}_k} \lambda_I = 1$ and since, $\sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) = 1$, it must hold that,

$$c_{\mathbf{m}} = \binom{n}{m_1, m_2, \dots, m_k}^{-1}, \quad \forall \mathbf{m} = (m_1, \dots, m_k) \in \mathcal{M},$$

□

A.2 Code Snippets and Explanations

```
from more_itertools \
import random_combination_with_replacement
k = 3; n = 10
true_labels = random_combination_with_replacement(range(3), 10)
#Output: (0, 0, 0, 1, 2, 2, 2, 2, 2, 2)
```

Listing A.1: Drawing true partition labels for n=10, into k=3 groups.

A.3 Additional figures and tables: Iris Dataset

Cluster	Feature	SW Statistic	p-value	Normal
0	Sepal Area	0.9607	0.9433	Yes
1	Sepal Area	0.7247	0.9666	Yes
2	Sepal Area	0.9660	0.8511	Yes
3	Sepal Area	0.9713	0.9526	Yes
0	Petal Area	0.4282	0.1767	Yes
1	Petal Area	1.5960	0.5612	Yes
2	Petal Area	0.1675	0.00002	No
3	Petal Area	0.2495	0.0405	Yes

Table A.1: Shapiro–Wilk test results for normality of sepal and petal area distributions across clusters. A significance level of $\alpha = 0.00625$ is used.

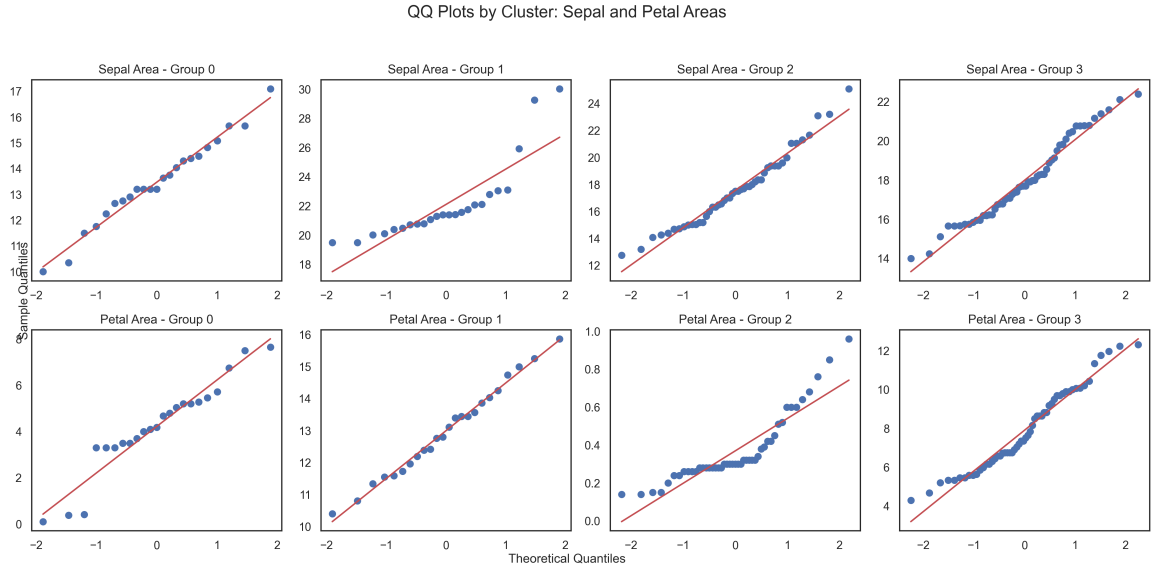


Figure A.1: Iris dataset: Assumptions check using QQ-plots, refer to Section 4.2.3.

Cluster	Chi-squared	p-value	Significant
0	67.618604	1.726806e-05	Yes
1	175.935256	0.000000e+00	Yes
2	343.219762	0.000000e+00	Yes
3	182.628098	2.220446e-16	Yes

Table A.2: Chi-squared test results for variance against $\sigma_0^2 = 1$ in each cluster

A.4 Code availability

The Python source code used to develop, test, and evaluate the Bayesian clustering methods presented in this thesis is publicly available at: [bayesiancl-repo]. The repository contains the following components:

- Complete implementation of the proposed Bayesian clustering algorithms.
- Scripts for synthetic data generation and benchmarking experiments.
- Evaluation metrics and comparison methods (e.g., KMeans, DBSCAN, ProdForm).
- Instructions to reproduce all figures and results from Chapters 3 and 4.

Configuration files are included to ensure full reproducibility of the experiments. Installation and usage instructions are provided in the accompanying README file.